TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

RRG(96) = 0
RRG(97) = 0
RRG(98) = 0
RRG(99) = PRG(69)
RRG(100)= 0
~~RRG(101)= 0~~
RRG(102)= 0
~~RRG(103)= 0~~
RRG(104)= PRG(4)
RRG(105)= 0
RRG(106)= 0
RRG(107)= 0
RRG(108)= 0
~~RRG(109)= 0~~
RRG(110)= PRG(10)
RRG(111)= 0
RRG(112)= 0
RRG(113)= 0
RRG(114)= 0
~~RRG(115)= 0~~
RRG(116)= PRG(16)
RRG(117)= 0
RRG(118)= 0
RRG(119)= 0
RRG(120)= 0
~~RRG(121)= PRG(20)~~
RRG(122)= PRG(26)
RRG(123)= PRG(32)
RRG(124)= PRG(47)
RRG(125)= PRG(48)
RRG(126)= PRG(77)
~~RRG(127)= PRG(59)~~
RRG(128)= PRG(60)
RRG(129)= PRG(75)
RRG(130)= PRG(64)
~~RRG(131)= PRG(24)~~
RRG(132)= PRG(30)
~~RRG(133)= PRG(36)~~
RRG(134)= PRG(37)
~~RRG(135)= PRG(38)~~
RRG(136)= PRG(39)
~~RRG(137)= PRG(41)~~
RRG(138)= PRG(42)
RRG(139)= PRG(44)
RRG(140)= PRG(45)
~~RRG(141)= PRG(49)~~
RRG(142)= PRG(43)
~~RRG(143)= PRG(50)~~
RRG(144)= PRG(51)
~~RRG(145)= PRG(52)~~
RRG(146)= PRG(55)

~~RRG(147)= PRG(56)~~
RRG(148)= PRG(58)
~~RRG(149)= PRG(61)~~
RRG(150)= PRG(62)
~~RRG(151)= PRG(65)~~
RRG(152)= PRG(66)
~~RRG(153)= PRG(67)~~
RRG(154)= PRG(68)
~~RRG(155)= PRG(70)~~
RRG(156)= PRG(71)
~~RRG(157)= PRG(72)~~
RRG(158)= PRG(73)
~~RRG(159)= PRG(74)~~
RRG(160)= PRG(46)
~~RRG(161)= PRG(78)~~
RRG(162)= 0

PRG(22)= PRG(22)*RTOD
~~PRG(23)= PRG(23)*RTOD~~
PRG(28)= PRG(28)*RTOD
PRG(29)= PRG(29)*RTOD
PRG(34)= PRG(34)*RTOD
PRG(35)= PRG(35)*RTOD
PRG(41)= PRG(41)*RTOD
~~PRG(64)=PRG(64)*RTOD~~
PRG(65)= PRG(65)*RTOD
~~PRG(75) = PRG(75)*RTOD~~
RRG(4) = PRG(22)
RRG(52)= PRG(23)
RRG(10)= PRG(28)
~~RRG(58)= PRG(29)~~
RRG(16)= PRG(34)
RRG(64)= PRG(35)
RRG(137)=PRG(41)
RRG(130)=PRG(64)
RRG(151)=PRG(65)
RRG(129)= PRG(75)

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
      IF(ICON(52).EQ.0) GO TO 515
      WRITE(14,25) PTIM
  515 CONTINUE
   25 FORMAT(6E15.7)
C     WRITE(6,2) PTIM,TSTT,DT,IRJU,IFAIL

      IF(ICON(52).EQ.0) GO TO 516
      WRITE(14,25) TIMC
  516 CONTINUE
C     WRITE(6,2) TIMC,TST,DT,IRJU,IFAIL

      IF(ICON(52).EQ.0) GO TO 533
      WRITE(14,26) (PR(I),I=1,30)
  533 CONTINUE
   26 FORMAT(6,(6E15.7)
      WRITE(6,5) (BT(I),PT(I),I=1,30)

      IF(ICON(52).EQ.0) GO TO 534
      WRITE(14,26) (PV(I),I=1,18)
  534 CONTINUE
      WRITE(6,5) (DV(I),PV(I),I=1,18)

      IF(ICON(15).GT.0) GO TO 544
      LINC = LINC + 14
      WRITE(6,4)
      IF(ICON(52).EQ.0) GO TO 541
      WRITE(14,26) (RM(I),I=1,72)
  541 CONTINUE

      IF(ICON(52).EQ.0) GO TO 554
      WRITE(14,26) (RG(I),I=1,162)
  554 CONTINUE
      WRITE(6,5) (DG(I),PG(I),I=1,NG)

      DIMENSION PT(30),PV(18),PM(54),PG(120),RG(162),RM(72)
```

RG(162),RM(72) added

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
PM( 1) = PM( 1)*GST
PM( 4) = PM( 4)*GST
PM( 6) = PM( 6)*GST
PM( 9) = PM( 9)*GST
PM(11) = PM(11)*GST
PM(16) = PM(16)*GST
PM(18) = PM(18)*GST
PM(19) = PM(19)*GST
PM(20) = PM(20)*GST
PM(22) = PM(22)*GST
PM(23) = PM(23)*GST
PM(24) = PM(24)*GST
PM(26) = PM(26)*GST
PM(27) = PM(27)*GST
PM(28) = PM(28)*RTOD
PM(29) = PM(29)*RTOD
PM(30) = PM(30)*RTOD
PM(31)= PM(31)*RTOD
PM(32) = PM(32)*GST
PM(33) = PM(33)*RTOD
PM(34) = PM(34)*RTOD
PM(35)= PM(35)*RTOD
PM(36) = PM(36)*RTOD
PM(37) = PM(37)*GST
PM(38)= PM(38)*RTOD
PM(39)= PM(39)*RTOD
PM(40)= PM(40)*RTOD
PM(41)= PM(41)*RTOD
PM(42) = PM(42)*GST
PM(46)= PM(46)*RTOD
PM(52) = PM(52)*RTOD
```

```
RM( 1) = PM( 1)
RM( 4) = PM( 4)
RM( 7) = PM( 6)
RM(10) = PM( 9)
RM(13) = PM(11)
RM(19) = PM(16)
RM(23) = PM(18)
RM(24) = PM(19)
RM(25) = PM(20)
RM(29) = PM(22)
RM(30) = PM(23)
RM(31) = PM(24)
RM(35) = PM(26)
RM(36) = PM(27)
RM(37) = PM(28)
RM(38) = PM(29)
RM(39) = PM(30)
RM(40) = PM(31)
RM(41) = PM(32)
RM(43) = PM(33)
RM(44) = PM(34)
RM(45) = PM(35)
RM(46) = PM(36)
RM(47) = PM(37)
RM(49) = PM(38)
RM(50) = PM(39)
RM(51) = PM(40)
RM(52) = PM(41)
RM(53) = PM(42)
RM(53) = PM(46)
RM(67) = PM(52)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
PG(120)=DVD-ZMTD
RG( 1) = PG(1)
RG( 2) = PG(2)
RG( 3) = PG(3)
RG( 4) = PG(4)
RG( 5) = PG(5)
RG( 6) = PG(6)
RG( 7) = PG(7)
RG( 8) = PG(8)
RG( 9) = PG(9)
RG(10) = PG(10)
RG(11) = PG(11)
RG(12) = PG(12)
RG(13) = PG(13)
RG(14) = PG(14)
RG(15) = PG(15)
RG(16) = PG(16)
RG(17) = PG(17)
RG(18) = PG(18)
RG(19) = PG(19)
RG(20) = PG(20)
RG(21) = PG(21)
RG(22) = PG(22)
RG(23) = PG(23)
RG(24) = PG(24)
RG(25) = PG(25)
RG(26) = PG(26)
RG(27) = PG(28)
RG(28) = PG(28)
RG(29) = PG(29)
RG(30) = PG(30)
RG(31) = PG(31)
RG(32) = PG(32)
RG(33) = PG(33)
RG(34) = PG(34)
RG(35) = PG(35)
RG(36) = PG(36)
RG(37) = PG(37)
RG(38) = PG(38)
RG(39) = PG(39)
RG(40) = PG(40)
RG(41) = PG(41)
RG(42) = PG(42)
RG(43) = PG(43)
RG(44) = PG(44)
RG(45) = PG(45)
INDGC = IAPFLG + 10*IGAFLG + 100*IACFLG + 1000*ISEQ
RINDGC = INDGC + 99.D5 + 0.000005
RG(46) = PG(46)
RG(47) = PG(47)
RG(48) = PG(48)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
      RG(49) = PG(49)
      RG(50) = PG(50)
      RG(51) = PG(51)
      RG(52) = PG(52)
      RG(53) = PG(53)
      RG(54) = PG(54)
      RG(55) = PG(55)
      RG(56) = PG(56)
      RG(57) = PG(57)
      RG(58) = PG(58)
      RG(59) = PG(59)
      RG(60) = PG(60)
      RG(61) = PG(61)
      RG(62) = PG(62)
      RG(63) = PG(63)
      RG(64) = PG(64)
      RG(65) = PG(65)
      IFTEMP = IALTU + 10*(IPUDFL+1) + 100*(IUPDFL+1) + 1000*
     1 JBSW + 10000*(IRCHFG+1)
      RG(66) = PG(66)
      RG(67) = PG(67)
      RG(68) = PG(68)
      RG(69) = PG(69)
      RG(70) = PG(70)
      RG(71) = PG(71)
      RG(72) = PG(72)
      RG(73) = PG(73)
      RG(74) = PG(74)
      RG(75) = PG(75)
      RG(76) = PG(76)
      RG(77) = PG(77)
      RG(78) = PG(78)
      RG(79) = PG(79)
      RG(80) = PG(80)
      RG(81) = PG(81)
      RG(82) = PG(82)
      RG(83) = PG(83)
      RG(84) = PG(84)
      RG(85) = PG(85)
      RG(86) = PG(86)
      RG(87) = PG(87)
      RG(88) = PG(88)
      RG(89) = PG(89)
      RG(90) = PG(90)
      RG(91) = PG(91)
      RG(92) = PG(92)
      RG(93) = PG(93)
      RG(94) = PG(94)
      RG(95) = PG(95)
      RG(96) = PG(96)
      RG(97) = PG(97)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
RG(98) = PG(98)
RG(99) = PG(99)
RG(100)= PG(100)
RG(101)= PG(101)
RG(102)= PG(102)
RG(103)= PG(103)
RG(104)= PG(104)
RG(105)= PG(105)
RG(106)= PG(106)
RG(107)= PG(107)
RG(108)= PG(108)
RG(109)= PG(109)
RG(110)= PG(110)
RG(111)= PG(111)
RG(112)= PG(112)
RG(113)= PG(113)
RG(114)= PG(114)
RG(115)= PG(115)
RG(116)= PG(116)
RG(117)= PG(117)
RG(118)= PG(118)
RG(119)= PG(119)
RG(120)= PG(120)
RG(121)= 0
RG(122)= 0
RG(123)= 0
RG(124)= 0
RG(125)= 0
RG(126)= 0
RG(127)= 0
RG(128)= 0
RG(129)= 0
RG(130)= 0
RG(131)= 0
RG(132)= 0
RG(133)= 0
RG(134)= 0
RG(135)= 0
RG(136)= 0
RG(137)= 0
RG(138)= 0
RG(139)= 0
RG(140)= 0
RG(141)= 0
RG(142)= 0
RG(143)= 0
RG(144)= 0
RG(145)= 0
RG(146)= 0
RG(147)= 0
RG(148)= 0
RG(149)= 0
RG(150)= 0
RG(151)= 0
RG(152)= 0
RG(153)= 0
RG(154)= 0
RG(155)= 0
RG(156)= 0
RG(157)= 0
RG(158)= 0
RG(159)= 0
RG(160)= 0
RG(161)= 0
RG(162)= 0
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
PG( 4)= PG( 4)*RTOD
PG(10)= PG(10)*RTOD
PG(16)= PG(16)*RTOD
PG(43)= PG(43)*RTOD
PG(49)= PG(49)*RTOD
PG(52) = PG(52)*RTOD
PG(55) = PG(55)*RTOD
PG(58) = PG(58)*RTOD
PG(64) = PG(64)*RTOD
PG(71)= PG(71)*RTOD
PG(77)= PG(77)*RTOD
PG(83)= PG(83)*RTOD
PG(88)= PG(88)*RTOD
PG(89)= PG(89)*RTOD
PG(90)= PG(90)*RTOD
PG(94)= PG(94)*RTOD
PG(95)= PG(95)*RTOD
PG(96)= PG(96)*RTOD
PG(97)= PG(97)*RTOD
PG(100)=PG(100)*RTOD
PG(101)=PG(101)*RTOD
PG(102)=PG(102)*RTOD
RG( 4) = PG( 4)
RG(10) = PG(10)
RG(16) = PG(16)
RG(43) = PG(43)
RG(49) = PG(49)
RG(52) = PG(52)
RG(55) = PG(55)
RG(58) = PG(58)
RG(64) = PG(64)
RG(71) = PG(71)
RG(77) = PG(77)
RG(83) = PG(83)
RG(88) = PG(88)
RG(89) = PG(89)
RG(90) = PG(90)
RG(94) = PG(94)
RG(95) = PG(95)
RG(96) = PG(96)
RG(97) = PG(97)
RG(100)= PG(100)
RG(101)= PG(101)
RG(102)= PG(102)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

PM(54) = WDACB
RM(1) = PM(1)
RM(2) = PM(2)
~~RM(3) = PM(3)~~
RM(4) = PM(4)
~~RM(5) = PM(5)~~
RM(6) = 0
~~RM(7) = PM(6)~~
RM(8) = PM(7)
~~RM(9) = PM(8)~~
RM(10) = PM(9)
~~RM(11) = PM(10)~~
RM(12) = 0
~~RM(13) = PM(11)~~
RM(14) = PM(12)
~~RM(15) = PM(13)~~
RM(16) = PM(14)
RM(17) = PM(15)
RM(18) = 0
RM(19) = PM(16)
RM(20) = 0
~~RM(21) = 0~~
RM(22) = PM(17)
RM(23) = PM(18)
RM(24) = PM(19)
RM(25) = PM(20)
RM(26) = 0
RM(27) = 0
RM(28) = PM(21)
RM(29) = PM(22)
RM(30) = PM(23)
RM(31) = PM(24)
RM(32) = 0
~~RM(33) = 0~~
RM(34) = PM(25)
RM(35) = PM(26)
RM(36) = PM(27)
RM(37) = PM(28)
RM(38) = PM(29)
RM(39) = PM(30)
RM(40) = PM(31)
RM(41) = PM(32)
RM(42) = 0
RM(43) = PM(33)
RM(44) = PM(34)
~~RM(45) = PM(35)~~
RM(46) = PM(36)
RM(47) = PM(37)
RM(48) = 0
RM(49) = PM(38)
RM(50) = PM(39)
RM(51) = PM(40)
RM(52) = PM(41)
RM(53) = PM(42)
RM(54) = 0
RM(55) = PM(43)
RM(56) = PM(44)
~~RM(57) = PM(45)~~
RM(58) = PM(46)
~~RM(59) = PM(47)~~
RM(60) = 0
RM(61) = PM(48)
RM(62) = PM(49)
~~RM(63) = 0~~
RM(64) = 0
~~RM(65) = PM(50)~~
RM(66) = PM(51)
~~RM(67) = PM(52)~~
RM(68) = PM(53)
~~RM(69) = PM(54)~~
RM(70) = 0
RM(71) = 0
RM(72) = 0

APPENDIX C
U70 PROGRAM .CSS (COMMAND SUBSTITUTE SYSTEM) SOURCE FILES

APPENDIX C

U70 PROGRAM .CSS (Command Substitute System) SOURCE FILES

The Perkin-Elmer 3220 computer on which the U70 program was run, has a development compiler which compiles rapidly but produces slow-running object code, and an optimizing compiler which compiles slowly but produces fast running code. The development compiler was used to compile the U70 program, since compile times rather than execution times were at a premium during the period of performance of this task. As stated in Section III,B, in order to further reduce compile times, each subroutine was assigned its own data file so that one could recompile one subroutine at a time rather than having to recompile the whole program each time a change was made in one of the subroutines.

There are two .CSS commands that can be used to actually carry out a compilation: U70CA and U70C. The U70CA command compiles all the Pershing II source files, while the U70C command is used to compile an individual subroutine. The specific job control sequence which initiates the U70 compilation is as follows:

```
U70 @1, @2, @3, @4, @5

@1  Input source file
@2  Object file name (defaults to @1.OBJ)
@3  List file name (defaults to @1.LST)
@4  Start option(s) separated with spaces (defaults to
    no options)
@5  Load size for F7D (defaults to 50K bytes).
```

A cross-reference listing can be obtained by including the command XREF in the @4 list of start options.

A preprocessing program called TRAJ must be run before the U70 program is ready to run. The TRAJ program is derived from equations resident in the Pershing Launch Computer, which is located on board the missile. Given the launch and target latitude, longitude, altitude, and the number of stages (Table C-1), it computes launch presets and transmits them to the portion of the program which simulates boost and terminal guidance. This, in turn, generates a data file containing estimates of the desired velocities and cutoff angles, estimates of flight and pitch-over times, and target offsets (Table C-2). The U70 program uses these outputs to make its own calculations, interpolating the TRAJ output tables.

Execution of the U70 program is initiated using the routines UST2 or UST3. These commands invoke .CSS files containing other commands to load the task, make logical unit assignments, and start the task.

The UST2 routine is used in order to make a full run (flight from launch to impact). The inputs to UST2 are obtained from the TRAJ programs and are used to build a new data file (logical unit 11). This file is used to create a new input tape in the U70 program, TAPE 11. The output from UST2.CSS is printed at time intervals whose value is input in Constant 271, or is printed at any point along the flight path where a discontinuity appears (such as the end of the boost phase, the apogee, etc.,).

The subroutines in the U70 program are linked using the .CSS command U70LINK. This command builds an overlaid executable task. For example, it ensures that copies of all of the subroutines called within a given subroutine are present within that subroutine.

TABLE C-1. TARGETING PROGRAM INPUTS

ITRS	=	210	Single Stage
		220	Two Stage
LONT	=		Longitude of the Target
LATDT	=		Latitude of the Target
HT	=		Altitude of the Target
LONL	=		Longitude of the Launch Site
LATDL	=		Latitude of the Launch Site
HL	=		Altitude of the Launch Site

TABLE C-2. GLOSSARY OF TARGETING PROGRAM OUTPUT VARIABLES

U70	TRAJECTORY OUTPUT	DEFINITION
140	GAMDE + DELGAM	Flight path angle desired before pitch-over
142	AZD	Flight azimuth measured from north
115	AZD	* Same value
114	LATDL	Launch latitude
116	LONL	Launch longitude
117	HL	Launch altitude
203	DLAT	Aimpoint offset latitude
205	(Unnamed)	Aimpoint offset altitude
206	LATDT	Target latitude
207	LONT	Target longitude
208	HT	Target altitude
219	GAMDE	GAMME desired at cut-off
220	TCO	Time of booster cut-off
302	TFF	Elasped time of re-entry vehicle freefall
310	TGAO	Elasped time before pitch-over
312	VRE	Required velocity for terminal phase
314	TF	Boost guidance flight time estimate
411	CHI	Normalized velocity loss
420	TAU	Initial delta value time impact
421	AXPU	Acceleration begins pull-up
412	DVP	Velocity control bias

U70LINK is structured so that it contains four overlays. The first overlay includes a subroutine called MINIT.OBJ, which presets all variables to zero. The second overlay includes ENPUT.OBJ, which creates a new input data tape. The third overlay includes all binary files appropriate to the boost phase of simulation. The fourth overlay includes all re-entry-vehicle binary code.

In addition, U70LINK creates a load map, U70-U70LINK.MAP, that is useful for debugging purposes. Control indicators are input to set flight options, and program constants are entered to control the initial conditions, multipliers, limits, errors, print controls, etc. Aerodynamic drag and atmospheric data are input through the tabular data portion of the input file. A list and description of these data inputs is given in Reference 1.

After a full run has been made, a restart can be created using the output from UST2.CSS and the UST3.CSS command. The restart procedure is a very useful tool in a study where new conditions are imposed on a missile at a certain point in flight. However, the restart could not be used for the nozzle deflection study, since a restart could not be made during the boost phase. Certain values which appeared in THRUST were functions of time and could not be re-initialized at the arbitrary time desired for a restart.

The values which are required to be input into the UST3.CSS data file of UST3.CSS are listed in Table C-3. Tables C-4, C-5, and C-6 list other U70 files used in the performance of this task for the benefit of anyone who might want to initiate U70 runs.

The procedure of outputting U70 flight data to tapes required that modifications be made to the subroutines BOUT (Boost Output) and RVOUT (Re-entry Vehicle Output). The desired output was written to logical unit 14 and a .CSS file was created, with logical unit 14 set to MAGØ, to write the data to the tapes.

TABLE C-3. GLOSSARY OF RESTART INPUT VARIABLES

U70	UST3, .CSS Input		Definition
101	TIME		Time desired for restart
102	XL	at time of restart	X component position in launch coordinate
103	YL	at time of restart	Y component position in launch coordinate
104	ZL	at time of restart	Z component position in launch coordinate
105	XLD	at time of restart	X component velocity in launch coordinate
106	YLD	at time of restart	Y component velocity in launch coordinate
107	ZLD	at time of restart	Z component velocity in launch coordinate
111	PHI	at time of restart	Euler angle
112	THETA	at time of restart	Euler angle
113	PSI	at time of restart	Euler angle
114	LATDL		Launch latitude
116	LONL		Launch longitude
117	HL		Launch altitude
140	GAMDE + DELGAM		Flight path angle desired at pitch-over
142	AZD		Azimuth
203	DLAT		Aimpoint offset in latitude
205	DALT		Aimpoint offset in altitude
206	LATDT		Target latitude
207	LONT		Target longitude
208	HT		Target altitude
219	GAMDE		Gamma desired at cut-off
220	TCO		Time of cut-off
314	TF		Boost guidance flight time estimate
420	TAU		Initial delta valve time impact
421	AXPU		Acceleration begins pull-up

TABLE C-4. U70 SOURCE FILES

ARD59D.FTN	QUAD.FTN
BEE.FTN	RVINIT.FTN
CONV.FTN	RVAPLT.FTN
FASTD.FTN	RVAERO.FTN
FASTC.FTN	SCS.FTN
SORTIT.FTN	GAUSS.FTN
SPEEDY.FTN	RANDU.FTN
WRTAPE.FTN	BEXEC.FTN
BINIT.FTN	THRUST.FTN
BGUID.FTN	TRIMA.FTN
FUNC.FTN	BOUT.FTN
VACRAN.FTN	BAPLT.FTN
FOLD.FTN	RATLIM.FTN
GRAPH.FTN	BAERO.FTN
NINIT.FTN	UST2.CSS
ENPUT.FTN	UST3.CSS
RVEXEC.FTN	UST4.CSS
STAT.FTN	U70LINK.CSS
RVOUT.FTN	U70C.CSS
RVGUID.FTN	

TABLE C-5. FILES USED FOR DEFLECTION STUDY

COB351.DAT	-	30 seconds, .5 degree pitch
COB352.DAT	-	30 seconds, 2.0 degree pitch
COB353.DAT	-	30 seconds, 7.6 degree pitch
COB354.DAT	-	49 seconds, .5 degree pitch
COB355.DAT	-	49 seconds, 2.0 degree pitch
COB356.DAT	-	49 seconds, 7.6 degree pitch
COB357.DAT	-	30 seconds, .5 degree yaw
COB358.DAT	-	30 seconds, 2.0 degree yaw
COB359.DAT	-	30 seconds, 7.6 degree yaw
COB360.DAT	-	49 seconds, .5 degree yaw
COB361.DAT	-	49 seconds, 2.0 degree yaw
COB362.DAT	-	49 seconds, 7.6 degree yaw

TABLE C-6. FILES USED FOR TRAJECTORY PROFILE

COB370.DAT
COB371.DAT
COB372.DAT
COB373.DAT
COB374.DAT
COB375.DAT
COB376.DAT
COB377.DAT
COB378.DAT

APPENDIX D
TRW PROGRAM FILE NAMES STORED ON DISK

TABLE D-1. SUBROUTINE FILE NAMES STORED ON DISK

TWAERO.FTN	TWMATRX1.FTN
TWAFC.FTN	TWMATRX2.FTN
TWALNON.FTN	TWMAXBT.FTN
TWAMC.FTN	TWNAVON.FTN
TWATMENT.FTN	TWNESTG.FTN
TWATMEXT.FTN	TWPWRON.FTN
TWATMOS.FTN	TWRADAR.FTN
TWAUX1.FTN	TWRCS.FTN
TWAUX2.FTN	TWRCSFM.FTN
TWBATT.FTN	TWRGU.FTN
TWBURST.FTN	TWRKUTTA.FTN
TWCMDPRC.FTN	TWRMMUL.FTN
TWCROSS.FTN	TWROTATE.FTN
TWDCU.FTN	TWRVAERO.FTN
TWDCUERR.FTN	TWRVBAS.FTN
TWDERIV.FTN	TWRVPTCH.FTN
TWDMNDC.FTN	TWRVROLL.FTN
TWDOT.FTN	TWRVYAW.FTN
TWDRAND.FTN	TWSAF.FTN
TWEBITS.FTN	TWSBITS.FTN
TWFLIGHT.FTN	TWSDPROC.FTN
TWFSICFU.FTN	TWSSICFU.FTN
TWFSISA.FTN	TWSSSCFU.FTN
TWFSSCFO.FTN	TWSTINIT.FTN
TWGRAV.FTN	TWTHRUST.FTN
TWGRN.FTN	TWTRNSF2.FTN
TWIMSALN.FTN	TWTRNSF3.FTN
TWIMSPLS.FTN	TWTRNSPO.FTN
TSIMSRSV.FTN	TWURN.FTN
TWIMSTST.FTN	TWVANES.FTN
TWINTERP.FTN	TWVMP.FTN
TWJDM.FTN3	TWWIND.FTN
TWMAGN.FTN	

TABLE D-2. FILES CONTAINED ON DISC FOR USER PROGRAM HANDLING

The following files are used:

DAVEF7.CSS - Used to compile each subroutine file.
TRWLINK.CSS - Used to link the TRW 6DOF simulator.
DCOMMENT.FTN - Contains comments which describe the function of each subroutine.
TRWRV.DAT - Contains re-entry aerodynamic data.
TRWSS.DAT - Contains single stage aerodynamic data.
TRWTAB.FTN - Used to read TRWRV.DAT and TRWSS.DAT and string them into a long one dimensional array. TRWTAB.CSS is used to run this program and TRWTAB.DAT contains the output of the TRWTAB.FTN.
BIG01R.DAT, BIG01I, BIG01D - Contain variables defined in the program and their storage locations in the BIG01 arrays, as specified by EQUIVALENCE statements.

The following files are used in the $INCLUDE statements to handle common blocks and equivalence statements.

TRW001.DAT
TRW002.DAT
TRW003.DAT
TRW004.DAT
TRW005.DAT
TRW006.DAT
TRW007.DAT
TRW008.DAT
TRW009.DAT
TRW0010.DAT
TRW0011.DAT
TRW0012.DAT
TRWCON.DAT
TRWCONST.DAT

APPENDIX E
TRW PROGRAM SUBROUTINE NAMES AND CALLING SEQUENCES

TABLE E-1. SUBROUTINE NAMES AND CALLING SEQUENCES

AERO	**DCU**	**FSISA**	**NEWSTG**	**SBITS**
RVAERO	INTERP	SBITS	TRNSF2	
AFC	SBITS			**SDPROC**
AMC	DCUERR	**FSSCFU**	**PWRON**	SBITS
		SBITS	TRNMAT	
AFC	**DCUERR**		TRNSPO	**SSICFU**
INTERP	INTERP	**GRAV**	STINIT	SBITS
			GRAV	
ALNON	**DERIV**	**GRN**	RMMUL	**SSSCFU**
MATRX1	VMP			SBITS
TRNMAT	AUX1	**IMSALN**	**RADAR**	
TRNSPO	NOZZLE	SBITS	TRNSF3	**STINIT**
RMMUL	THRUST	RMMUL		RMMUL
	JDM	TRNMAT	**RCS**	
	AUX1	TRNSPO		**THRUST**
AMC	AERO		**RCSFM**	INTERP
INTERP		**IMSPLS**	CROSS	
	DMNDC	RMMUL		**TRNSF2**
ATMENT	SAF		**RGU**	
	SDPROC	**IMSRSV**		**TRNSF3**
ATMEXT	BATT		**RKUTTA**	
	FSICFU	**IMSTST**	DERIV	**TRNSPO**
ATMOS	FSISA	SBITS		
	FSSCFU		**RMMUL**	**URN**
AUX1	SSICFU	**INTERP**		
MATRX1	SSSCFU		**ROTATE**	**VANES**
GRAV	BURST	**JDM**		
RMMUL	SBITS		**RAVERO**	**VMP**
MAGN	CMDPRC	**MAGN**	RVBAS	INTERP
ATMOS			RVROLL	
	DOT	**MATRX1**	RVPTCH	**WIND**
AUX2	MAGN	TRNMAT	RVYAW	INTERP
MAGN		ROTATE		
MATRX2	**DRAND**	RMMUL	**RVBAS**	
RMMUL		TRNSPO	INTERP	
	EBITS			
BATT		**MATRX2**	**RVPTCH**	
SBITS	**FLIGHT**	TRNMAT	INTERP	
	RKUTTA			
BITS	AUX1	**MAXBT**	**RVROLL**	
	AUX2		INTERP	
BURST	IMSALN	**NAVON**		
	IMSPLS	TRNMAT	**RVYAW**	
CMDPRC	IMSRSV	TRNSPO	INTERP	
SBITS	RGU	MATRX1		
		STINIT	**SAF**	
CROSS	**FSICFU**	GRAV	SBITS	
UNITV	SBITS	RMMUL		

APPENDIX F

ORGANIZATION OF THE INTERP(INTERPOLATION) DATA TABLES

APPENDIX F

ORGANIZATION OF THE INTERP(INTERPOLATION) DATA TABLES

An interpolation routine called INTERP forms the backbone of the TRW program's aerodynamic simulator, and utilizes a large set of multidimensional tables to generate its results. The multidimensional interpolation process is straightforward and is described with the aid of the following example.

Suppose that we have a table of pressures:

TABLE F-1. AERODYNAMIC PRESSURES

0	0.1	0.3	0.5 (bars)

and a table of temperatures,

TABLE F-2. AERODYNAMIC TEMPERATURES

1000°	1300°	1600°

The pressure and temperature are considerd to be independent variables. Associated with these independent variables is a table (Table F-3) of aerodynamic coefficients, C:

TABLE F-3. AERODYNAMIC COEFFICIENTS

0.30021	0.30106	0.30028
0.30021	0.30176	0.30057
0.30021	0.30268	0.30081
0.30022	0.30345	0.30117

Now suppose that we are given pressure, P, = 0.3306 (bars) and temperature, T, = 1116°, and we want to interpolate to find the aerodynamic coefficients that correspond to these values of pressure and temperature. First, the interpolation routine tests the input pressure value P, against the table of pressures and determines that the input value P, = 0.3306 lies between 0.3 and 0.5. The location of the upper number - the "0.5" - is read out and stored in a variable which might be labeled "PINDEX" for Pressure INDEX. Since 0.5 is the fourth element in this independent variable table, the number stored in PINDEX is a "4", and it is a pointer to the interval in the independent variable table in which 0.3306 is located.

Next, the same operations are carried out for the temperature, using the temperature table. Since the temperature lies between 1000° and 1300°, the location of the 1300° value - location #2 - is stored in TINDEX (Temperature INDEX) to designate the proper interpolation interval.

Finally, the two index values, 4 and 2, are used to locate the interval around the 4th row and 2nd column of the dependent variable table of aerodynamic coefficients (Table F-3). Then the pressure and temperature to be evaluated, P = 0.3306 bars and T = 1116°, are used to carry out the actual interpolation, in order to obtain the interpolated values of the aerodynamic coefficients.

In the listing of the INTERP subroutine's data output, the following commands appear:

```
BEGIN
INDEPENDENT_TABLE OUTPUT VARIABLE/
                INPUT VARIABLE
DEPENDENT_TABLE OUTPUT VARIABLE/
            INPUT VARIABLE, INPUT VARIABLE
END.
```

At first, these were thought to be VAX 11/780 commands. Later, it was determined that the TRW program contains its own data input processor and logical analyzer, which reads and interprets input data tapes, including the above statements. Apparently, these table definitions are used only for printer output formatting. Using the example given above, these statements would take the following form:

```
BEGIN
INDEPENDENT_TABLE PINDEX/PRESUR
INDEPENDENT_TABLE TINDEX/TEMP
DEPENDENT_TABLE AEROCO/PINDEX, TINDEX
END
```

When the results are printed out, AEROCO, standing for "Aerodynamic Coefficients", would be printed at the top of the column listing the aerodynamic coefficient values which INTERP generates by table interpolation.

APPENDIX G
FLOWCHART OF AN EARLY MAIN PROGRAM

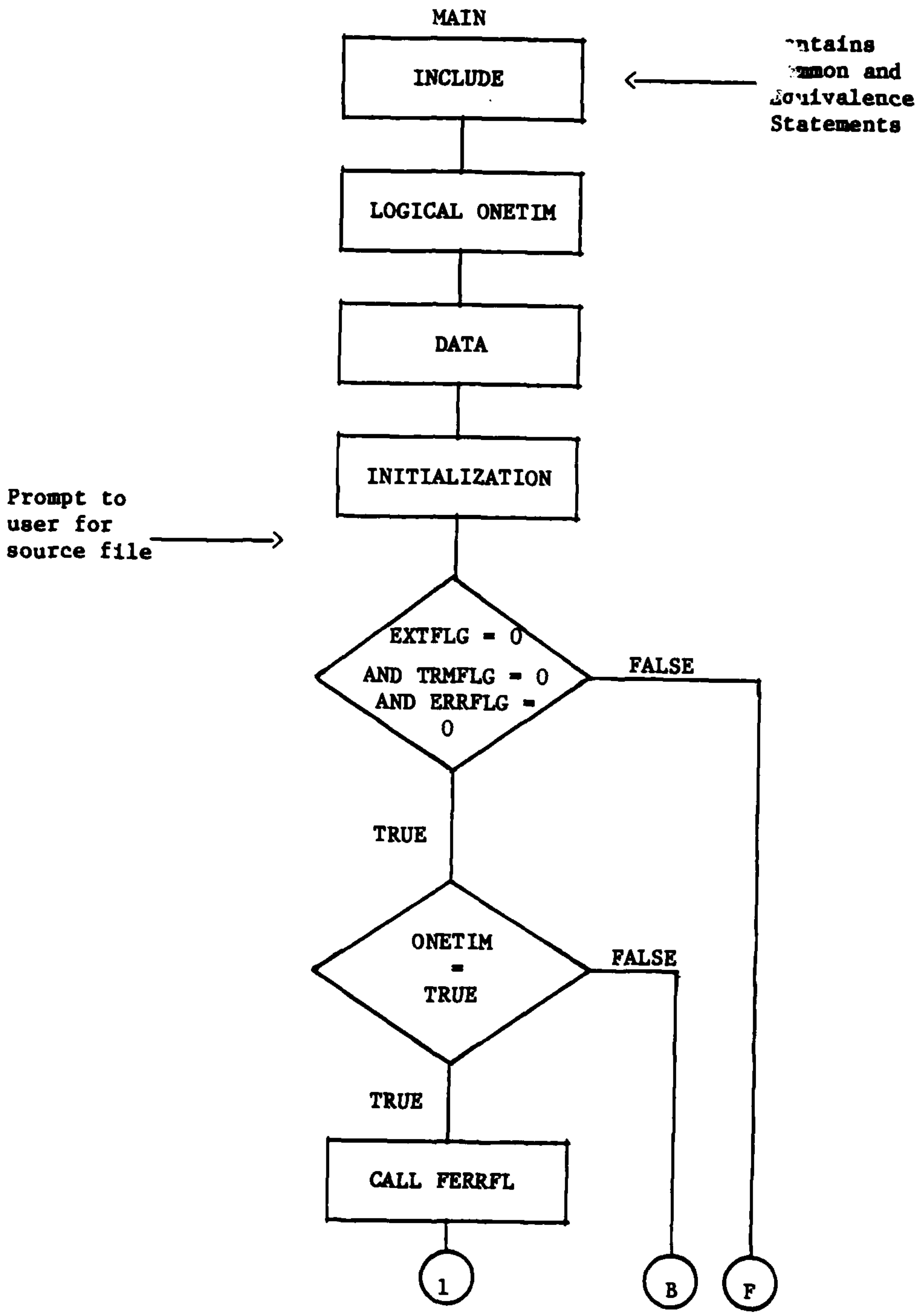

Figure G-1. FLowchart of Early MAIN Program.

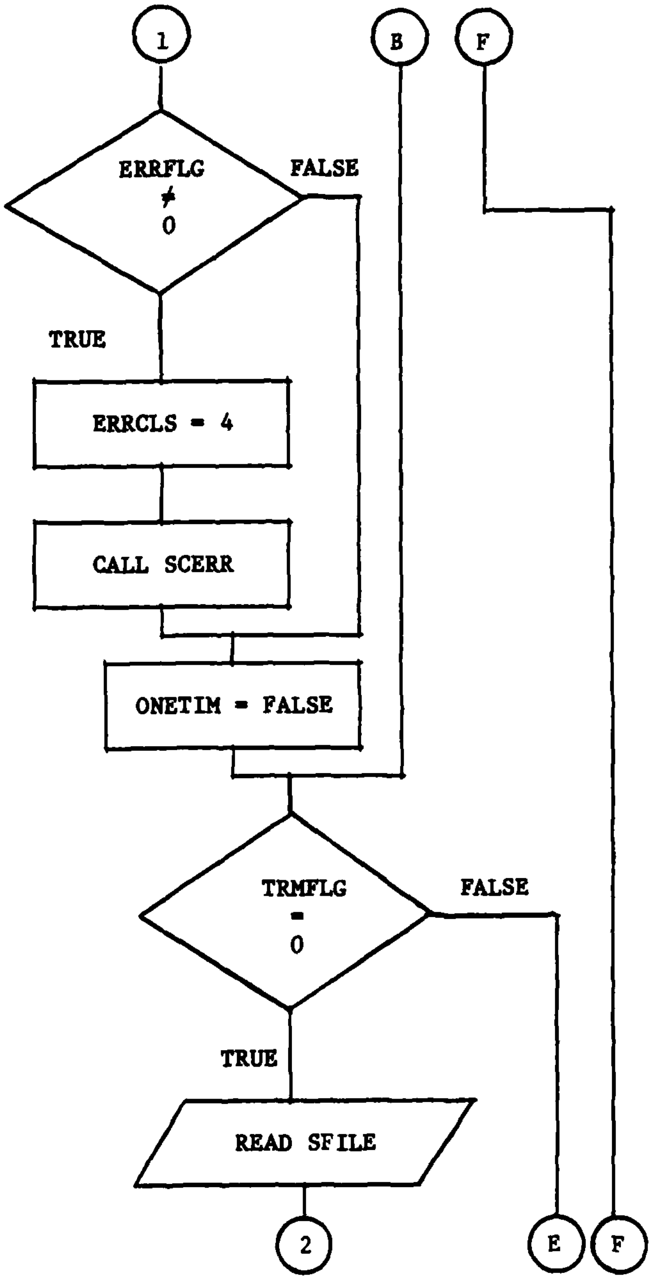

Figure G-1. Flowchart of Early MAIN Program (Continued).

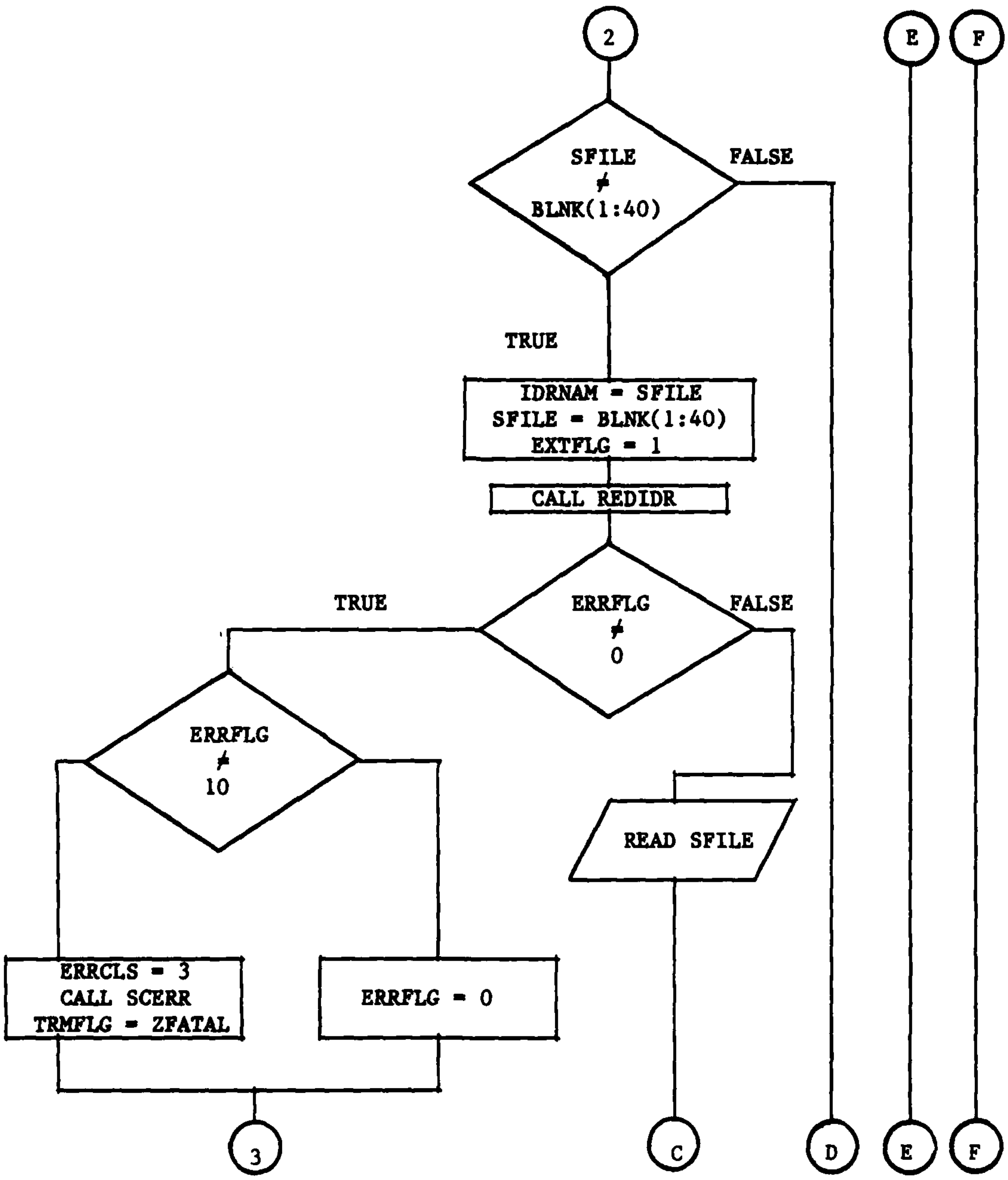

Figure G-1. Flowchart of Early MAIN Program (Continued).

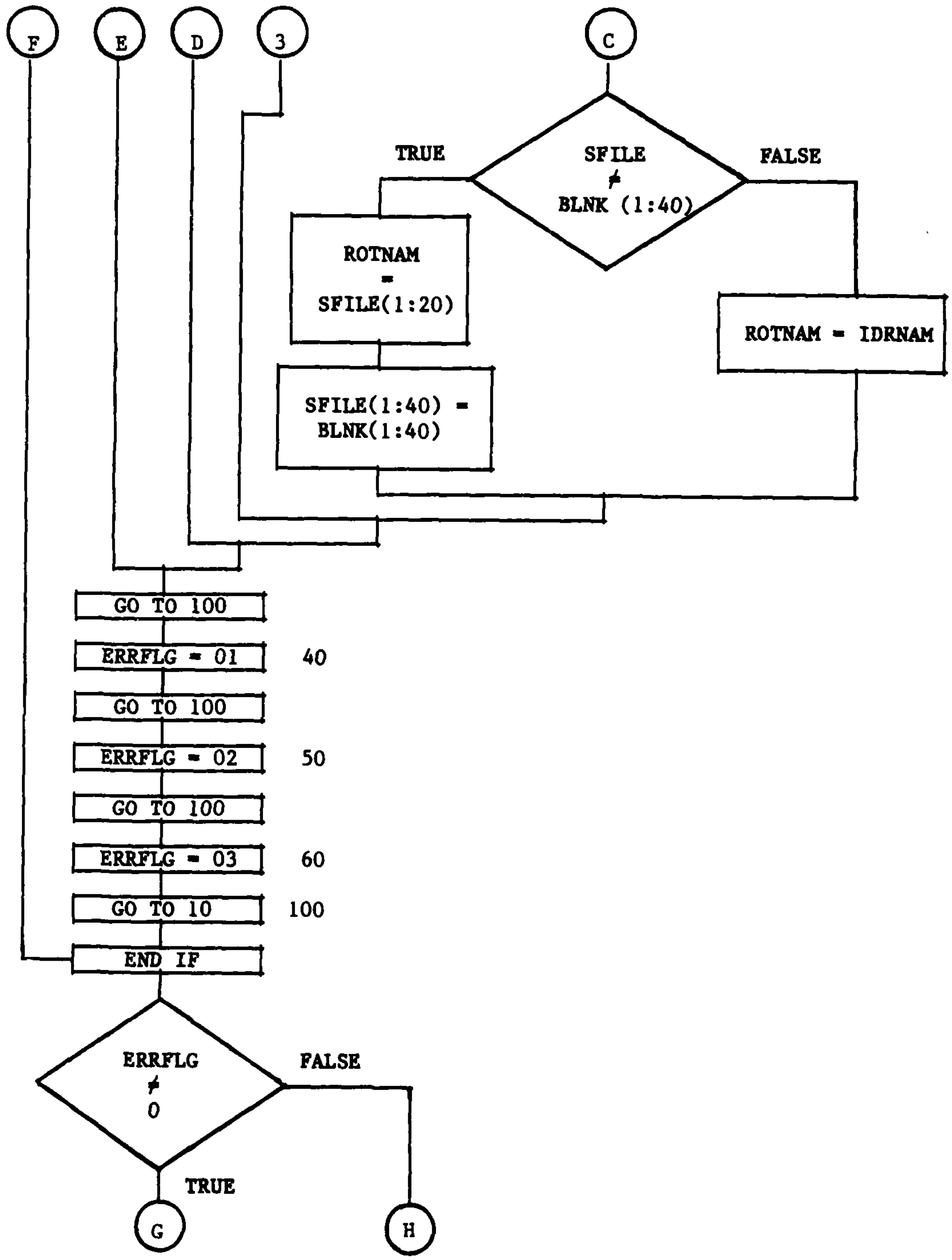

Figure G-1. Flowchart of Early MAIN Program (Continued).

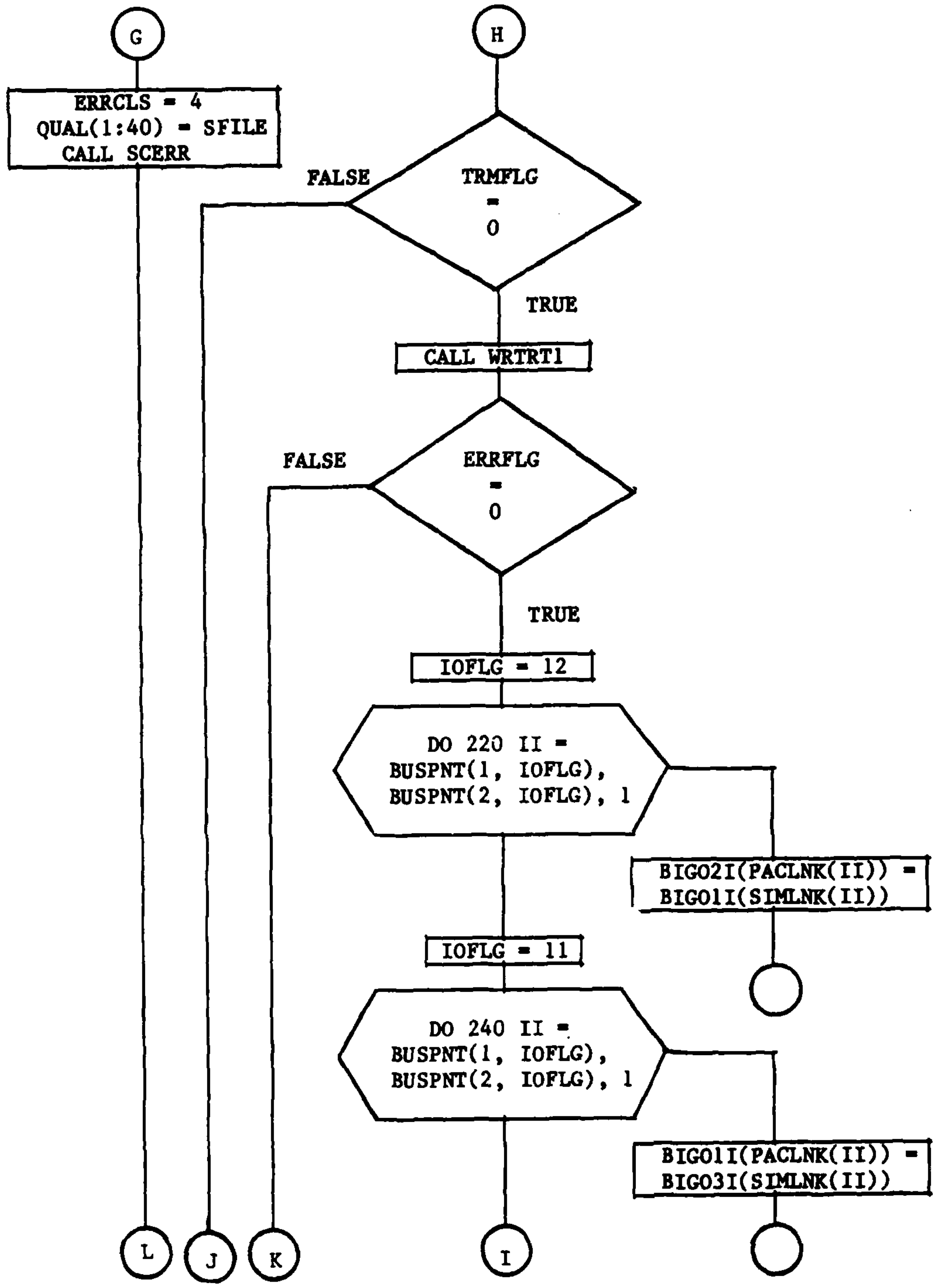

Figure G-1. Flowchart of Early MAIN Program (Continued).

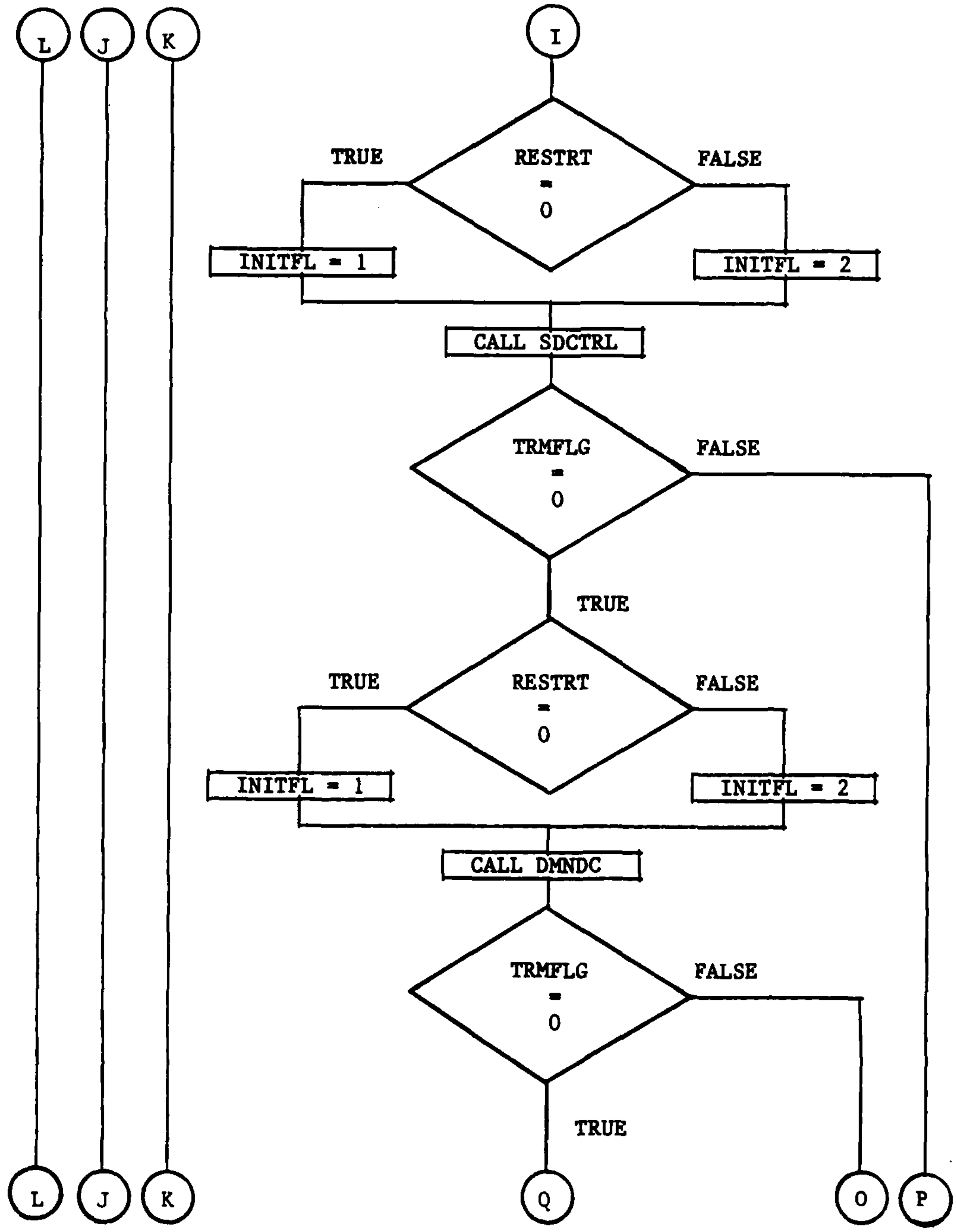

Figure G-1. Flowchart of Early MAIN Program (Continued).

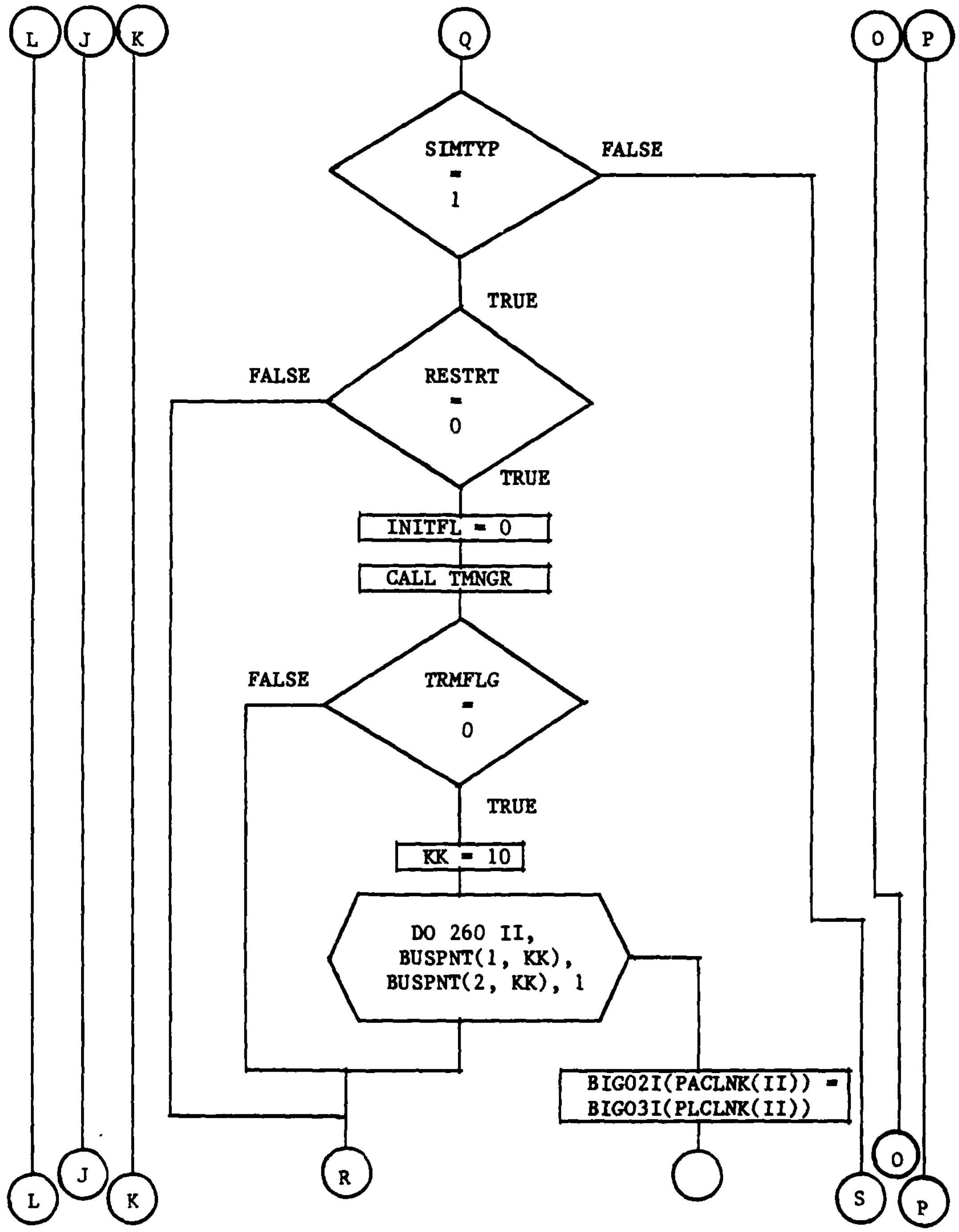

Figure G-1. Flowchart of Early MAIN Program (Continued).

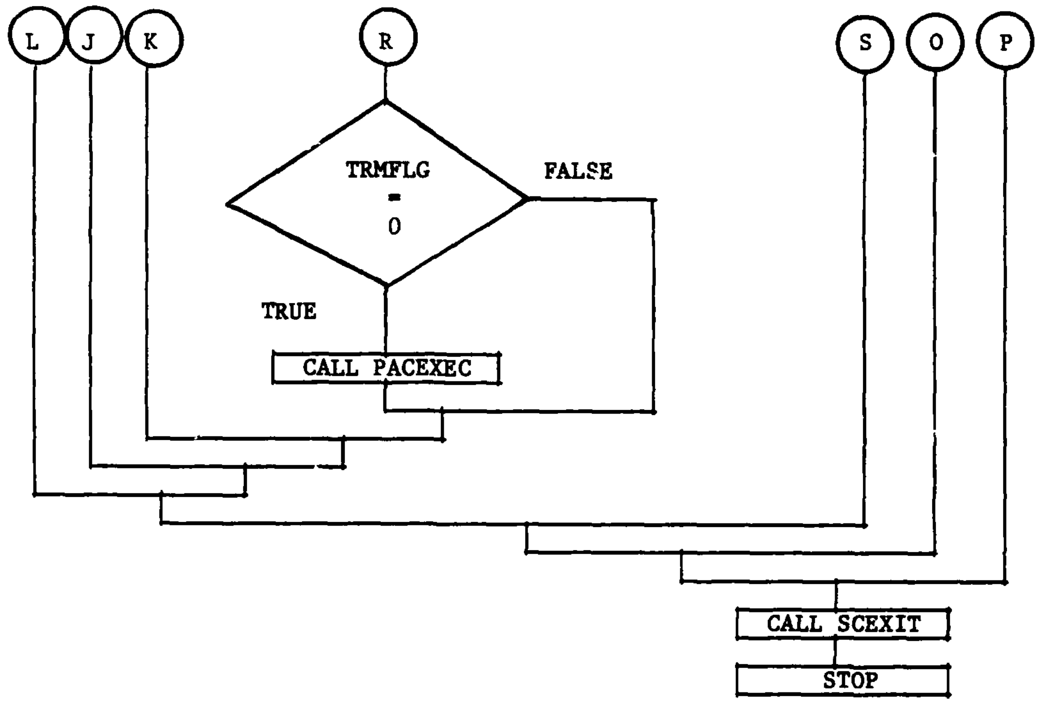

Figure G-1. Flowchart of Early MAIN Program (Continued).

APPENDIX H
FLOWCHART OF THE SUBROUTINE SDCTRL
(SIX-DEGREE-OF-FREEDOM CONTROL) PROGRAM

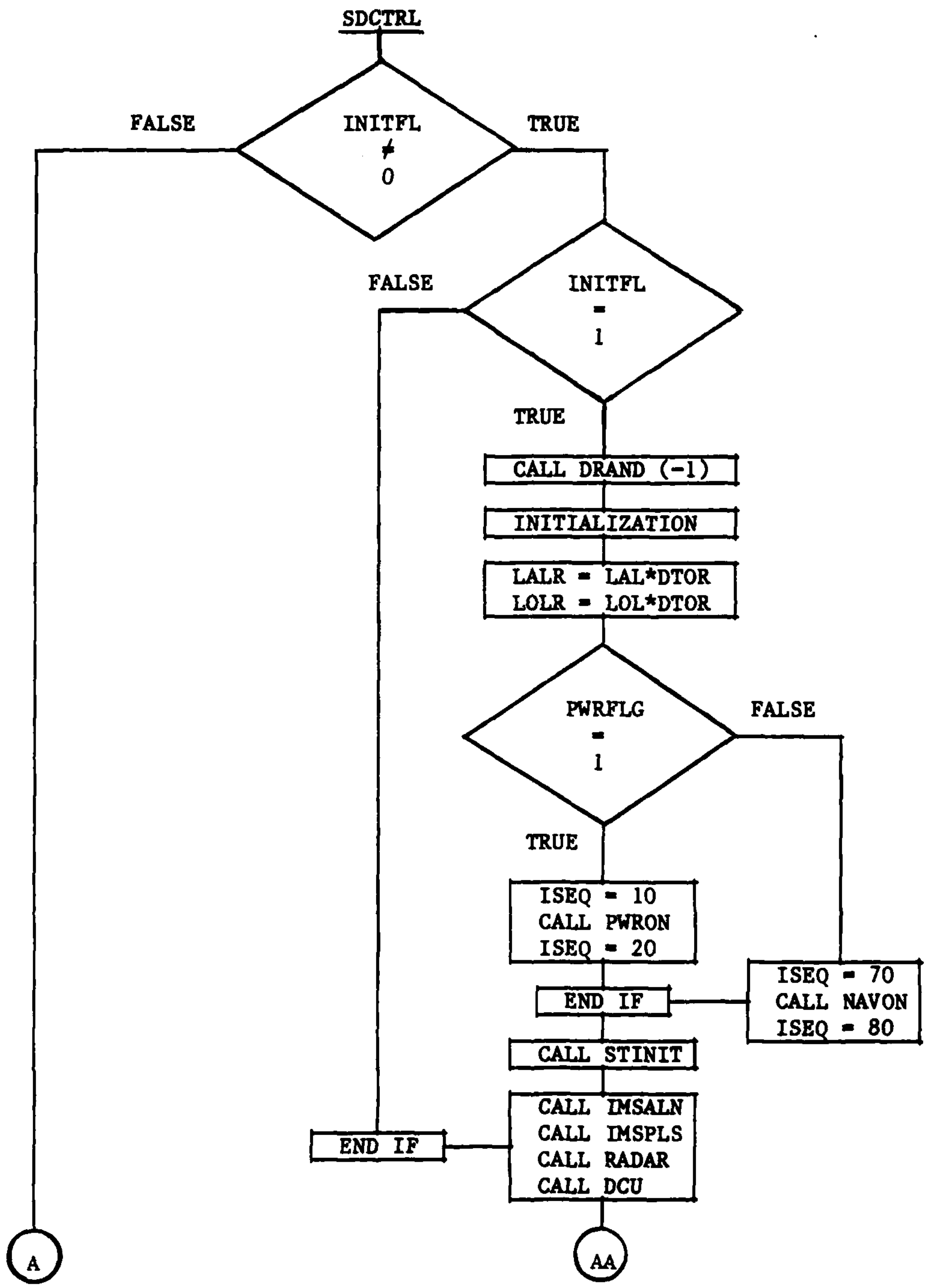

Figure H-1. Flowchart of SDCTRL Program.

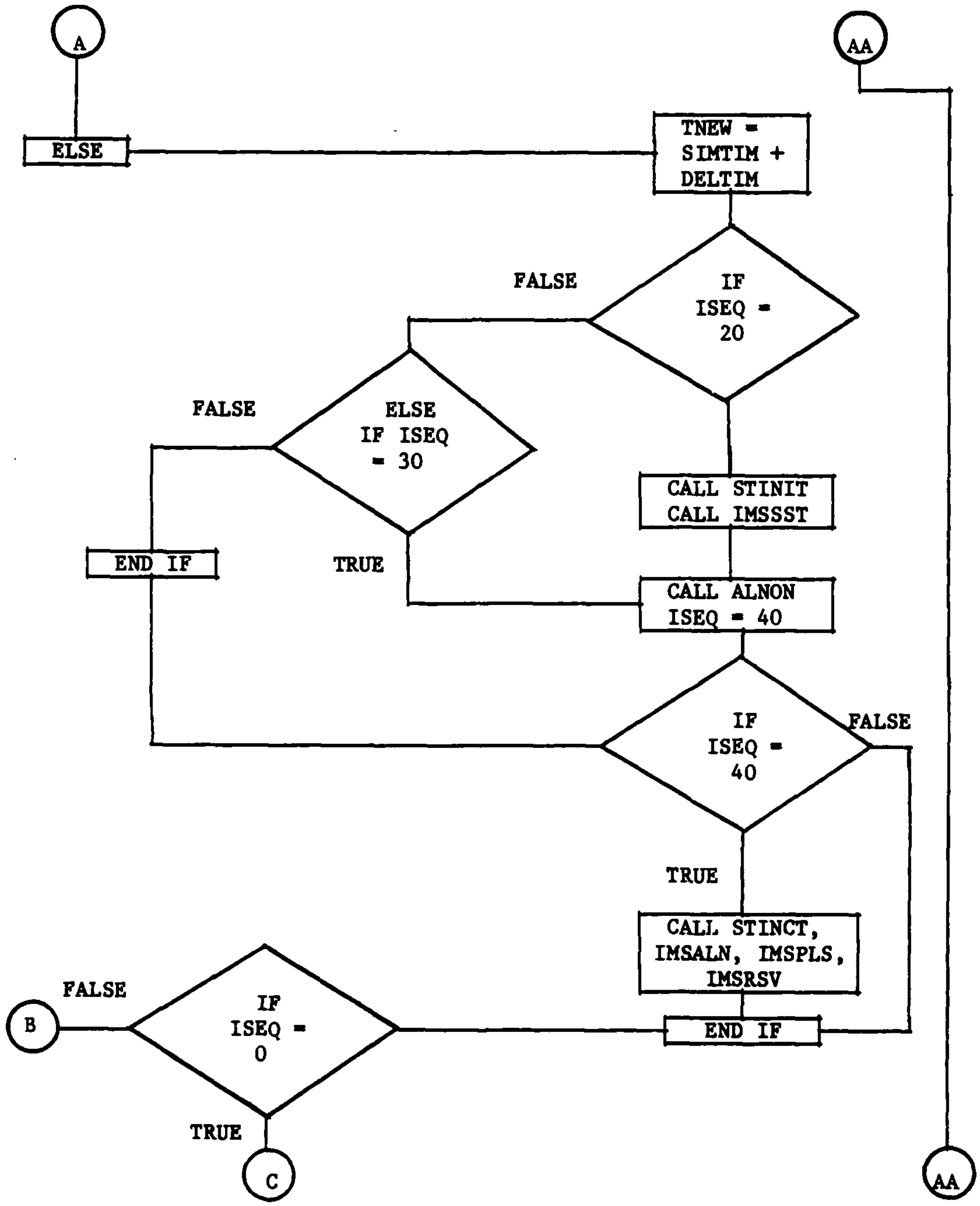

Figure H-1. Flowchart of SDCTRL Program (Continued).

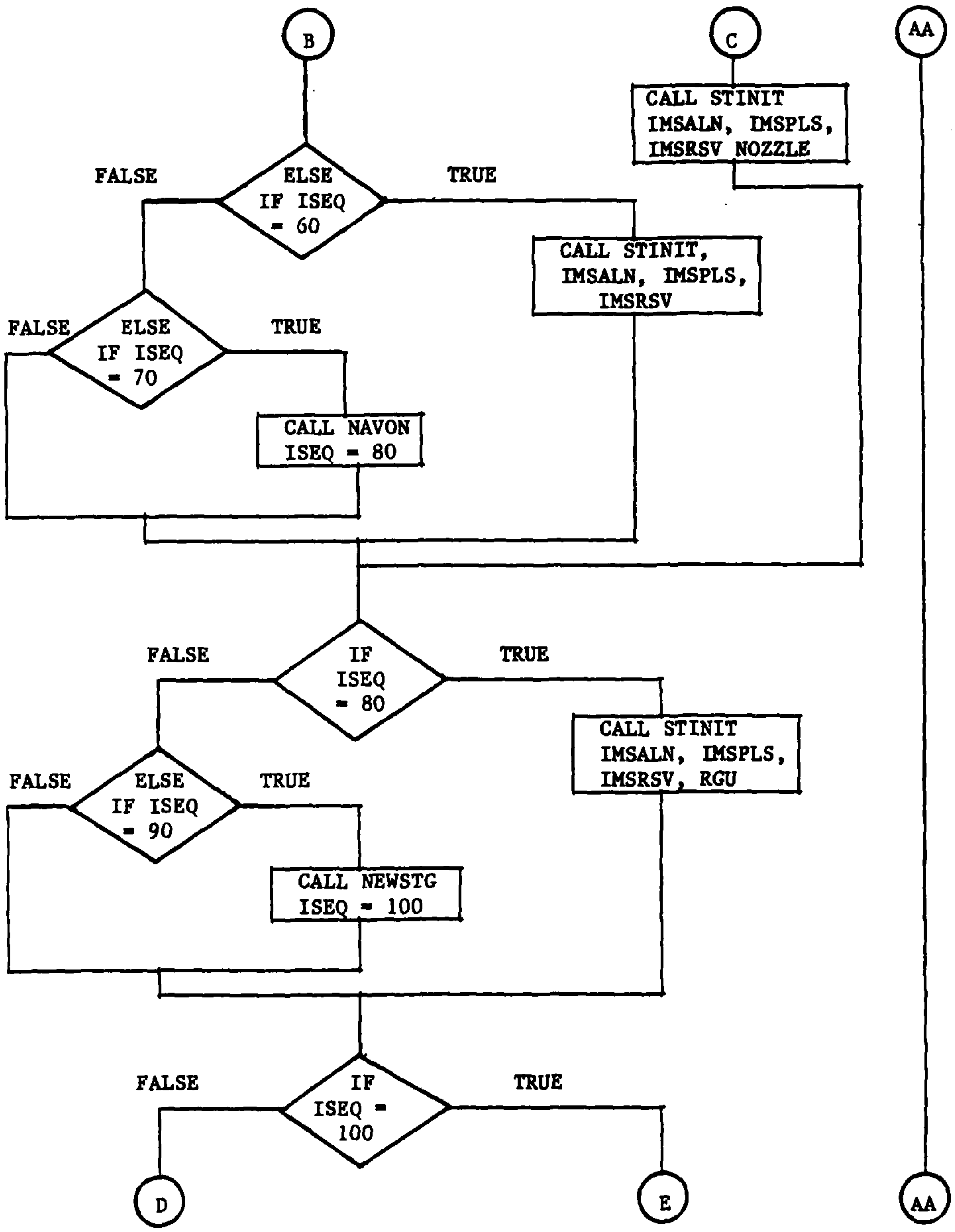

Figure H-1. Flowchart of SDCTRL Program (Continued).

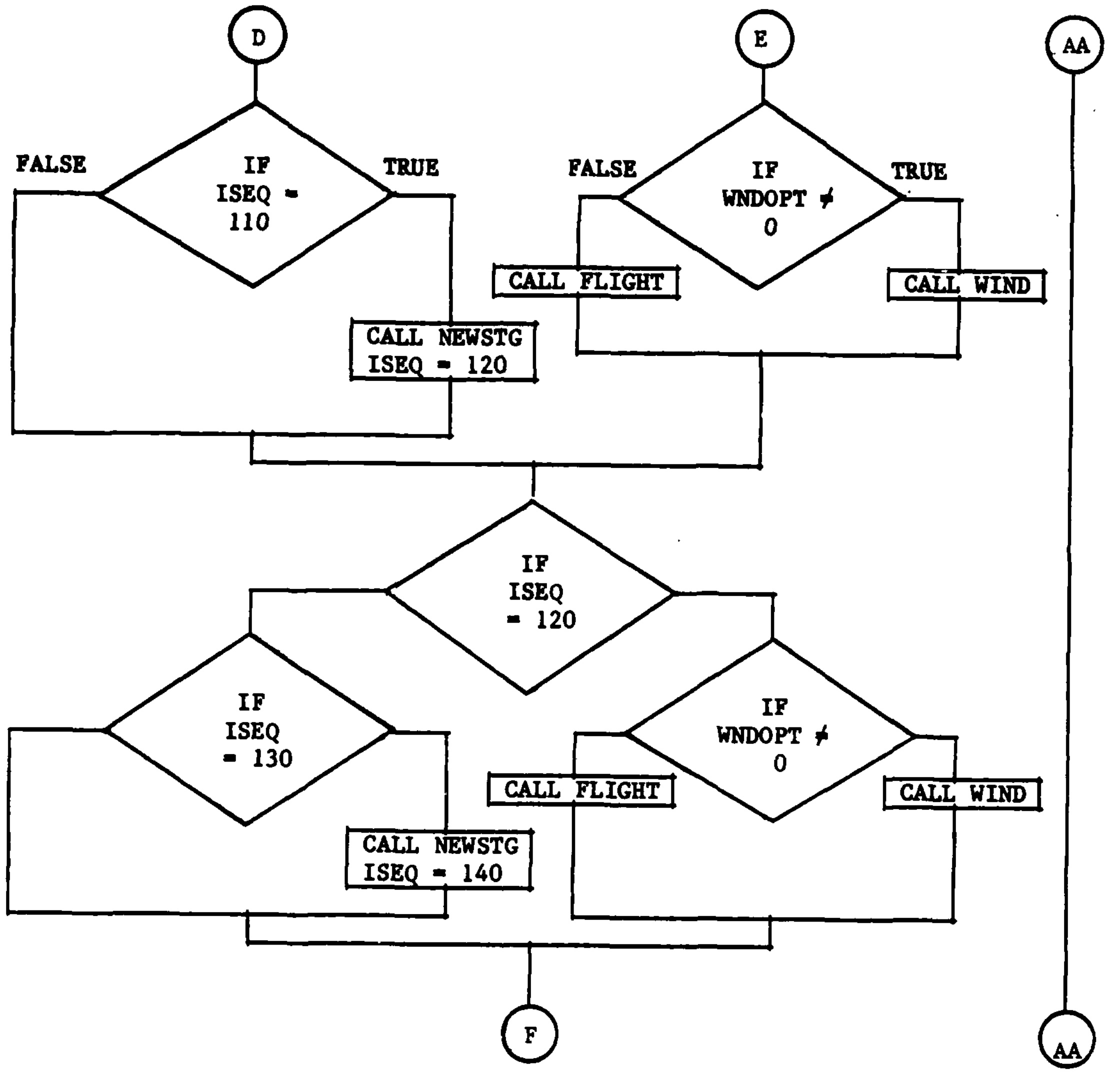

Figure H-1. Flowchart of SDCTRL Program (Continued).

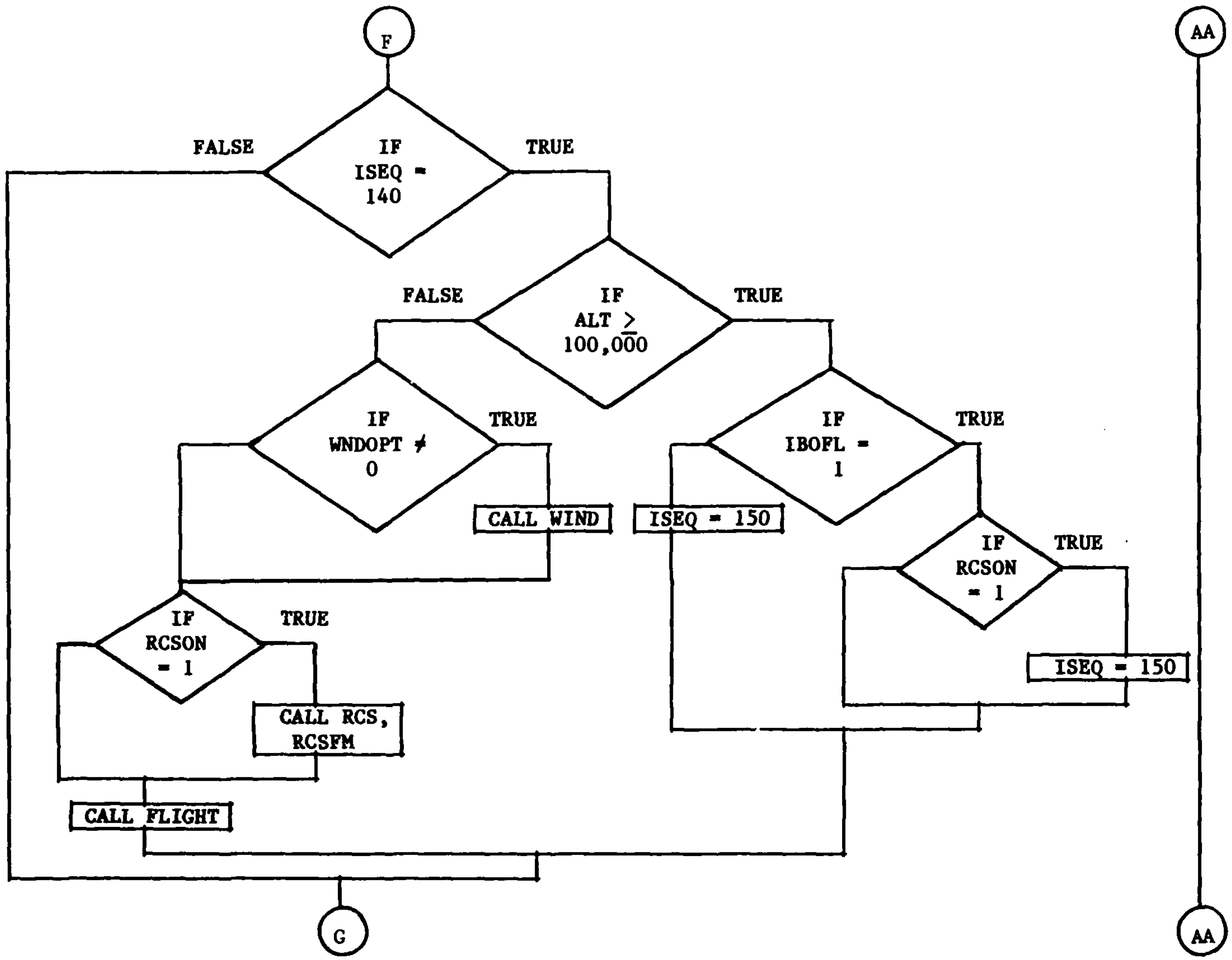

Figure H-1. Flowchart of SDCTRL Program (Continued).

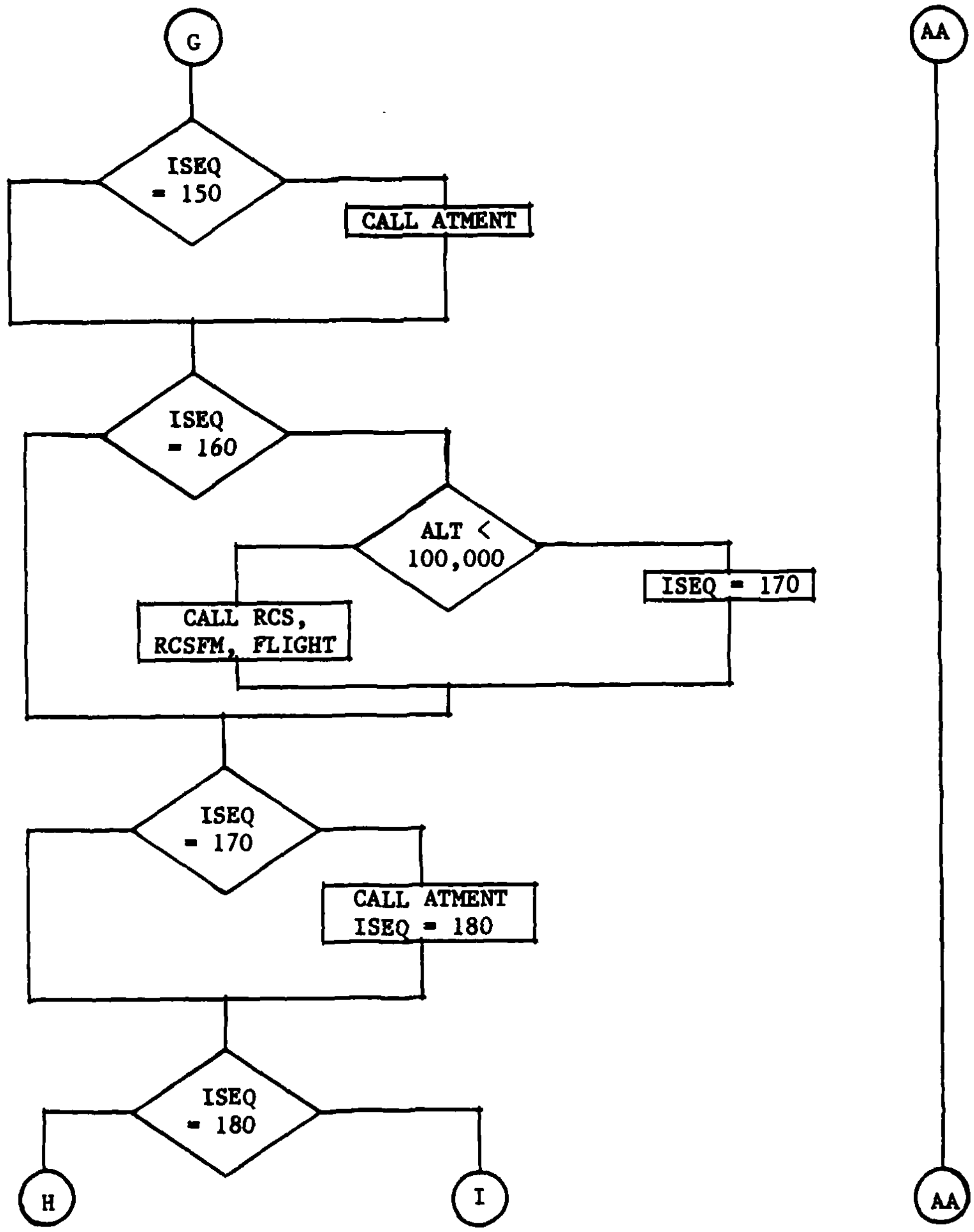

Figure H-1. Flowchart of SDCTRL Program (Continued).

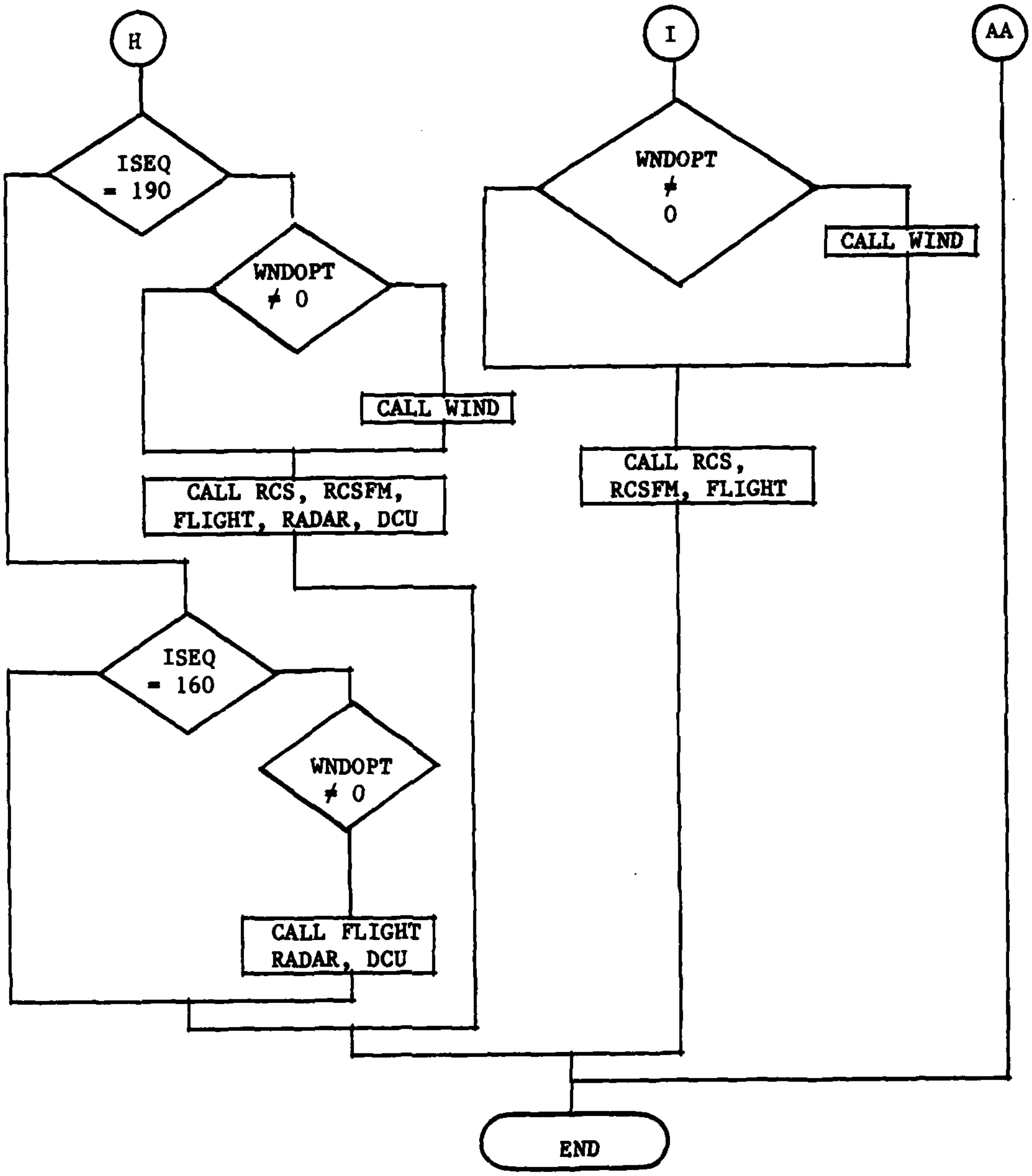

Figure H-1. Flowchart of SDCTRL Program (Continued).

APPENDIX I
TRW PROGRAM OUTPUT VARIABLES

TABLE I-1. TRW PROGRAM OUTPUT VARIABLES

```
AEARTH  BIG01R(  1)EQUATORIAL EARTH RADIUS
ALT     BIG01R(  2)ALTITUDE
AOT     BIG01R(  3)ALTITUDE OF TARGET
DCM I   BIG01R(  4)INITIAL CM OFFSET
DELTIM  BIG01R(  7)INCREMENT FROM LAST PASS TO BE ADDED SIMTIM
DER     BIG01R(  8)STATE VARIABLE DERIVATIVES
DLFSV   BIG01R( 20)FIRST STAGE VANE DEFL. ANGLES
DLRVV   BIG01R( 24)RV VANE DEFL. ANGLES
EFTODF  BIG01R( 28)EARTH-FIXED TO DESIRED TRANSFORMATION MATRIX
EFTOMF  BIG01R( 37)EARTH-FIXED TO MISSILE TRANSFORMATION MATRIX
EFTOND  BIG01R( 46)EARTH-FIXED TO LOCAL NED TRANSFORMATION MATRIX
FRCSM   BIG01R( 55)RCS THRUST VECTOR IN MISSILE FRAME
GE      BIG01R( 60)GRAVITY VECTOR IN EARTH-FIXED FRAME
GM      BIG01R( 63)
HEAD    BIG01R( 66)AZIMUTH OF MISSILE LONGITUDINAL AXIS ON EL
LAL     BIG01R( 74)LATITUDE OF LAUNCH
LALR    BIG01R( 75)LATITUDE OF LAUNCH IN RADIANS
LOL     BIG01R( 76)LONGITUDE OF LAUNCH
LOLR    BIG01R( 77)LONGITUDE OF LAUNCH IN RADIANS
MDOT    BIG01R( 78)MAIN ENGINE MASS FLOW RATE
MFTOEF  BIG01R( 79)MISSILE TO EARTH FIXED TRANSFORMATION MATRIX
MJDM    BIG01R( 88)JET DAMPING MOMENT VECTOR IN MISSILE FRAME
MRCSM   BIG01R( 91)RCS MOMENT VECTOR IN MISSILE FRAME
MTM     BIG01R( 94)PROPULSIVE THRUST MOMENT VECTOR MISSILE FRAME
NOZSTA  BIG01R( 97)NOZZLE PIVOT POINT LOCATION-X COMPONENT
P       BIG01R( 99)ROLL, PITCH, AND YAW BODY RATES-P, Q, AND R
PR      BIG01R(102)ATMOSPHERIC PRESSURE
PSI     BIG01R(103)EULER ANGLES-PLATFORM TO MISSILE FRAME
RE      BIG01R(109)OBLATE EARTH RADIUS
TFRVS   BIG01R(117)TIME FROM RV SEPARATION
TM      BIG01R(119)BASIC PROPULSIVE THRUST VECTOR IN MISSILE FRAME
TNEW    BIG01R(122)TIME AT END OF CURRENT STEP (SIMTIM + DELTIM)
TOL     BIG01R(123)TIME OF LIFT-OFF
TORVS   BIG01R(124)TIME OF RV SEPARATION
TOSI    BIG01R(125)TIME OF SECOND STAGE IGNITION
U       BIG01R(126)VELOCITY IN MISSILE FRAME
WE      BIG01R(129)EARTH ROTATION RATE
XCM STA BIG01R(130)MISSILE TRANSIENT CM LOCATION
XE      BIG01R(131)POSITION IN EARTH-FIXED FRAME
XLE     BIG01R(134)LAUNCH POSITION VECTOR IN EARTH-FIXED FRAME
XTE     BIG01R(137)TARGET POSITION VECTOR IN EARTH-FIXED FRAME
ALFDOT  BIG01R(140)TIME DERIVATIVE OF ALPHAM
ALPHAM  BIG01R(141)PITCH PLANE COMPONENT OF ANGLE OF ATTACK
ALPHAT  BIG01R(142)TOTAL ANGLE OF ATTACK
AMCDLX  BIG01R(143)AERO. MOMENT COEFFICIENT TRANSFER DISTANCE
AMCSTA  BIG01R(144)AERO. MOMENT COEFFICIENT REFERENCE LOCATION
AOL     BIG01R(145)ALTITUDE OF LAUNCH
AOL     BIG01R(145)ALTITUDE OF LAUNCH
AREF    BIG01R(146)AERO. REFERENCE AREA
BETAM   BIG01R(147)YAW PLANE COMPONENT OF ANGLE OF ATTACK
BETDOT  BIG01R(148)TIME DERIVATIVE OF BETAM
```

TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

```
CAT     BIG01R(149)TOTAL AERO. AXIAL FORCE COEFFICIENT
CN      BIG01R(150)BASIC NORMAL FORCE COEFFICIENT
CN      BIG01R(150)PARAMETER FOR ACCELERATION LIMIT
CNSTAT  BIG01R(151)TOTAL STATIC NORMAL FORCE COEFFICIENT
CNT     BIG01R(152)TOTAL AERO. NORMAL FORCE COEFFICIENT
CPMT    BIG01R(153)TOTAL AERO. PITCHING MOMENT COEFFICIENT
CRMT    BIG01R(154)TOTAL AERO. ROLLING MOMENT COEFFICIENT
CS      BIG01R(155)SPEED OF SOUND
CY      BIG01R(156)BASIC SIDE FORCE COEFFICIENT
CYMT    BIG01R(157)TOTAL AERO. YAWING MOMENT COEFFICIENT
CYSTAT  BIG01R(158)TOTAL STATIC SIDE FORCE COEFFICIENT
CYT     BIG01R(159)TOTAL AERO. SIDE FORCE COEFFICIENT
DELRV   BIG01R(160)RV VANE DELFECTION USED IN INTERPOLATION
DELTAP  BIG01R(161)EQUIVALENT ROLL CONTROL VANE DEFL.
DELTAQ  BIG01R(162)EQUIVALENT PTICH CONTROL VANE DEFL.
DELTAR  BIG01R(163)EQUIVALENT YAW CONTROL VANE DEFL.
DLFSVP  BIG01R(164)FIRST STAGE ROLL CONTROL VANE DEFL.
DXDRL   BIG01R(165)TOTAL AERO. MOMENT COEF. TRANSFER DISTANCE
DYNPR   BIG01R(166)DYNAMIC PRESSURE
ELIPTE  BIG01R(167)ELLIPICITY OF EARTH
EPSAB   BIG01R(168)BIAS ERRORS IN TOTAL AERO. COEFFICIENTS
EPSAF   BIG01R(174)FRACTIONAL ERRORS IN TOTAL AERO. COEFFICIENTS
EPSCMB  BIG01R(180)BIAS ERROR IN LONGITUDINAL CM POSITION
EPSIB   BIG01R(181)BIAS ERRORS IN MOMENTS AND PRODUCTS OF INERTIA
EPSIF   BIG01R(187) ERRORS IN MOMENTS AND PRODUCTS OF INERTIA
EPSMB   BIG01R(193)MASS BIAS ERROR
EPSVEB  BIG01R(194)BIAS ERROR IN MOMENT COEF. DUE TO VANE DEFL.
EPSVEF  BIG01R(197)ERROR IN MOMENT COEF. DUE TO VANE DEFL.
FAM     BIG01R(200)AERO. FORCE VECTOR IN MISSILE FRAME
FWM     BIG01R(203)MISSILE WEIGHT VECTOR IN MISSILE FRAME
ITEN    BIG01R(206)INERTIA TENSOR
ITENDT  BIG01R(215)TIME RATE OF CHANGE OF MOMENTS OF INERTIA
LREF    BIG01R(219)AERO. REFERENCE LENGTH
MACH    BIG01R(220)MACH NUMBER
MAM     BIG01R(221)AERO. MOMENTS ABOUT MISSILE AXES
MASS    BIG01R(224)BOOST MASS ESTIMATE
MASS    BIG01R(224)VEHICLE MASS
MASSGG  BIG01R(225)MASS EXPELLED BY GAS GENERATOR
MDGG    BIG01R(226)GAS GENERATOR MASS FLOW RATE
PHIAP   BIG01R(228)AERO. ROLL ANGLE MEASURED IN A/P FRAME
PHIPR   BIG01R(229)AERO. ROLL ANGLE
PHIRV   BIG01R(230)MODIFIED AERO. ROLL ANGLE FOR INTERPOLATION
RHO     BIG01R(231)ATMOSPHERIC DENSITY
RLO2V   BIG01R(232)LREF / (2.0 * VAT)
RN      BIG01R(233)REYNOLDS NUMBER
RNTRNS  BIG01R(234)BOUNDARY LAYER TRANSITION REYNOLDS NUMBER
SUMFM   BIG01R(235)SUMMATION OF FORCES IN MISSILE FRAME
SUMMM   BIG01R(238)SUMMATION OF MOMENTS IN MISSILE FRAME
TFL     BIG01R(241)TIME FROM LIFT-OFF
TFSI    BIG01R(242)TIME FROM SECOND STAGE IGNITION
TOLD    BIG01R(243)
```

TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

```
VEF     BIG01R(244)MISSILE INERTIAL VEL. W.R.T. EARTH-FIXED FRAME
VAT     BIG01R(247)MAGNITUDE OF MISSILE VELOCITY W.R.T. AIR
VE      BIG01R(248)MISSILE EARTH-RELATIVE VELOCITY
VE      BIG01R(248)EARTH RELATIVE VELOCITY VECTOR
VI      BIG01R(249)MISSILE INERTIAL VELOCITY
VISC    BIG01R(250)COEFFICIENT OF VISCOSITY
VLAT    BIG01R(251)VEHICLE LATITUDE
VLON    BIG01R(252)VEHICLE LONGITUDE
WINDE   BIG01R(253)WIND VECTOR IN EARTH-FIXED FRAME
XEWE    BIG01R(2?1)EARTH RELATIVE VELOCITY COMPONENTS
AEXIT   BIG01R(309)NOZZLE EXIT AREA
D1P     BIG01R(310)FIRST STAGE NOZZLE PITCH COMMAND
D1Y     BIG01R(311)FIRST STAGE NOZZLE YAW COMMAND
D2P     BIG01R(312)SECOND STAGE NOZZLE PITCH COMMAND
D2Y     BIG01R(313)SECOND STAGE NOZZLE YAW COMMAND
D31     BIG01R(314)RV VANE 1 COMMAND
D32     BIG01R(315)RV VANE 2 COMMAND
D33     BIG01R(316)RV VANE 3 COMMAND
D34     BIG01R(317)RV VANE 4 COMMAND
DCJ     BIG01R(31?)DIRECTION COSINES OF THE RCS THRUST VECTORS
DLFSVD  BIG01R(342)FIRST STAGE VANE ANGULAR RATES
DLNP    BIG01R(346)NOZZLE PITCH ANGLE
DLNPD   BIG01R(347)NOZZLE PITCH ANGULAR RATE
DLNY    BIG01R(348)NOZZLE YAW ANGLE
DLNYD   BIG01R(349)NOZZLE YAW ANGULAR RATE
DLRVVD  BIG01R(350)RV VANE ANGULAR RATES
DRC     BIG01R(354)FIRST STAGE VANE ROLL COMMAND
DTRCS   BIG01R(355)RCS JET TIME DELAY
EPSTBS  BIG01R(356)THRUST BIAS ERROR
EPSTRC  BIG01R(357)FRACTIONAL ERRORS IN RCS THRUST MAGNITUDE
EPSTGF  BIG01R(3?5)FRACTIONAL ERROR IN THRUST
FRCS    BIG01R(356)NOMINAL MAGNITUDE OF RCS THRUST FOR ITH JET
GMU     BIG01R(374)GRAVITY PARAMETER
J2      BIG01R(375)SECOND ORDER ZONAL HARMONIC
J3      BIG01R(376)THIRD ORDER ZONAL HARMONIC
J4      BIG01R(377)FOURTH ORDER ZONAL HARMONIC
LAMNZ   BIG01R(37?)
LJD     BIG01R(379)CHARACTERISTIC JET DAMPING LENGTH POSITION
MAJ     BIG01R(3?2)MOMENT ARM VECTORS OF THE RCS JETS (8 JETS)
NLIMFS  BIG01R(406)FIRST STAGE NOZZLE DEFL. ANGLE LIMIT
NLIMSS  BIG01R(407)SECOND STAGE NOZZLE DEFL. ANGLE LIMIT
PHINZ   BIG01R(408)
PSLV    BIG01R(409)SEA LEVEL PRESSURE
RCSDLX  BIG01R(410)RCS MOMENT ARM COMPONENT ALONG X AXIS
RCSSTA  BIG01R(411)RCS LOCATION-X COMPONENT
TNOZ    BIG01R(420)TIME AT WHICH LAST NOZZLE ANGLES COMPUTED
TVANES  BIG01R(421)TIME AT WHICH LAST VANE ANGLES COMPUTED
VLIMFS  BIG01R(422)FIRST STAGE VANE DEFL. ANGLE LIMIT
```

TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

```
VLIMRV BIG01R(423)RV VANE DEFL. ANGLE LIMIT
WFSV   BIG01R(424)FIRST STAGE VANE NATURAL FREQUENCY
WNP    BIG01R(425)NOZZLE PITCH NATURAL FREQUENCY
WNY    BIG01R(426)NOZZLE YAW NATURAL FREQUENCY
WRVV   BIG01R(427)RV VANE NATURAL FREQUENCY
XNOZ   BIG01R(428)INITIAL NOZZLE POSITION WITH RESPECT TO CM
ZETANP BIG01R(431)NOZZLE PITCH DAMPING FACTOR
ZETANY BIG01R(432)NOZZLE YAW DAMPING FACTOR
ZETFSV BIG01R(433)FIRST STAGE VANE DAMPING FACTOR
ZETRVV BIG01R(434)RV VANE DAMPING FACTOR
ABIAS  BIG01R(453)ACCELEROMETER BIASES
AGP    BIG01R(456)GRAVITATIONAL ACCELERATION VECTOR
AGPOLD BIG01R(459)PREVIOUS GRAVITATIONAL ACCELERATION VECTOR
ARO    BIG01R(462)AZIMUTH (YAW)RESOLVER OFFSET
ARO    BIG01R(462)AZIMUTH (YAW)RESOLVER OFFSET
DDYX   BIG01R(465)Y TO X NONORTHOGONALITY PRESET
DDYX   BIG01R(465)Y TO X NONORTHOGONALITY PRESET
DELYX  BIG01R(466)NONORTHOGONALITY BETWEEN Y AND X ACCELEROMETERS
DELZX  BIG01R(467)NONORTHOGONALITY BETWEEN Z AND X ACCELEROMETERS
DELZY  BIG01R(468)NONORTHOGONALITY BETWEEN Z AND Y ACCELEROMETERS
DF     BIG01R(469)GYRO RESTRAINT DRIFTS-X, Y, AND Z
DI     BIG01R(472)INPUT MASS UNBALANCE DRIFTS-X, Y, AND Z
DOZ    BIG01R(475)Z GYRO ACCELERATION ALONG X CLUSTER
DO     BIG01R(476)OUTPUT MASS UNBALANCE DRIFTS-X, Y, AND Z
EIMS   BIG01R(479)EULER ANGLES-TRUE IMS PLATFORM TO PLATFORM
EPSBS  BIG01R(482)FRACTIONAL ERROR ACCELEROMETER BIAS ESTIMATE
EPSCBG BIG01R(485)GIMBAL ANGLE BIASES FOR COSINE RESOLVERS
EPSCFG BIG01R(488)FRACTIONAL ERROR IN IMS COSINE RESOLVER MEASURE
EPSFD  BIG01R(491)ERROR IN PLATFORM FIXED DRIFT ESTIMATES
EPSGD  BIG01R(494)FRACTIONAL ERROR IN G-SENSITIVE DRIFT PARTIALS
EPSNSF BIG01R(497)ERROR IN NEG. ACCELEROMETER SCALE FACTOR
EPSORT BIG01R(500)ERROR IN ACCELEROMETER NONORTHOGONALTIY
EPSPSF BIG01R(503)ERROR IN POS. ACCELEROMETER SCALE FACTOR
EPSSBG BIG01R(506)GIMBAL ANGLE BIASES FOR SINE RESOLVERS
EPSSFG BIG01R(509)FRACTIONAL ERROR IN IMS SINE RESOLVER MEASURE
EPSTD  BIG01R(512)ERROR IN TORQUER SLEW RATES
IMSTIM BIG01R(516)IMS TEST TIMER
IRTOIF BIG01R(517)INERTIAL REFERENCE TO INERTIAL EARTH TRANS.
IRTOPF BIG01R(526)INERTIAL REFERENCE TO PLATFORM TRANS. MATRIX
KI     BIG01R(535)ACCELEROMETER LOW GAIN SCALE ERRORS-XYZ
KIH    BIG01R(538)ACCELEROMETER HIGH GAIN SCALE ERRORS-XYZ
KNHSF  BIG01R(541)NOMINAL NEGATIVE HIGH GAIN ACCELEROMETER SCALE
KNLSF  BIG01R(542)NOMINAL NEGATIVE LOW GAIN ACCELEROMETER SCALE
KO     BIG01R(543)ACCELEROMETER LOW GAIN BIASES-X,Y,AND Z
KOH    BIG01R(546)ACCELEROMETER HIGH GAIN BIASES-X, Y, AND Z
KPHSF  BIG01R(549)NOMINAL POSITIVE HIGH GAIN ACCELEROMETER SCALE
KPLSF  BIG01R(550)NOMINAL POSITIVE LOW GAIN ACC. SCALE FACTOR
KS     BIG01R(551)ACCELEROMETER LOW GAIN SYMMETRY ERRORS-XYZ
KSH    BIG01R(554)ACCELEROMETER HIGH GAIN SYMMETRY ERRORS-XYZ
KT     BIG01R(557)GYRO TORQUER SCALE FACTOR RATIO
LAT    BIG01R(560)LATITUDE OF TARGET
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TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

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LOT     BIG01R(561)LONGITUDE OF TARGET
MFTOIR  BIG01R(562)MISSILE TO INERTIAL REFERENCE TRANS. MATRIX
MFTOND  BIG01R(571)MISSILE TO LOCAL NED TRANS. MATRIX
MFTOPF  BIG01R(580)MISSILE TO PLATFORM TRANS. MATRIX
PFTOIR  BIG01R(596)PLATFORM TO INERTIAL REFERENCE TRANS. MATRIX
PRO     BIG01R(605)PITCH RESOLVER OFFSET
PSIDTP  BIG01R(606)PREVIOUS VALUE OF PRE-LAUNCH EULER ANGLE RATES
RCPCH   BIG01R(609)COSINE IMS PITCH
RCROL   BIG01R(610)COSINE IMS ROLL
RCYAW   BIG01R(611)COSINE IMS YAW
REL     BIG01R(612)LAUNCH LOCAL EARTH RADIUS
RET     BIG01R(613)TARGET LOCAL EARTH RADIUS
RHG     BIG01R(614)HIGH GAIN ACCELEROMETER DRIFT-X AND Y
RLG     BIG01R(616)LOW GAIN ACCELEROMETER DRIFT-X AND Y
RRO     BIG01R(618)ROLL RESOLVER OFFSET
RSPCH   BIG01R(619)SINE IMS PITCH
RSROL   BIG01R(620)SINE IMS ROLL
RSYAW   BIG01R(621)SINE IMS YAW
SIGDLV  BIG01R(622)STANDARD DEVIATION OF DELTA V SENSED BY IMS
SIGRD   BIG01R(623)DEVIATING PLATFORM DRIFT FROM RANDOM ERRORS
SIGRGA  BIG01R(624)STANDARD DEVIATION OF IMS RESOLVER MEASURE
VIR     BIG01R(626)MISSILE VELOCITY IN INERTIAL REFERENCE FRAME
VP      BIG01R(629)MISSILE VELOCITY IN PLATFORM FRAME
VPOLD   BIG01R(632)PREVIOUS VELOCITY VECTOR SENSED BY IMS
ALTP    BIG01R(635)PREVIOUS ALT FROM DCU ERROR MODEL
APHA    BIG01R(636)TRANSFER FUNCTION -ROLL ANTENNA ANGLE
APSA    BIG01R(637)TRANSFER FUNCTION -YAW ANTENNA ANGLE
ATHA    BIG01R(638)TRANS. FUNCTION -PITCH ANTENNA ANGLE
BPE     BIG01R(639)EAST ERROR PARAMETER FROM DCU ERROR MODEL
BPHA    BIG01R(640)TRANS. FUNCTION -ROLL ANTENNA ANGLE
BPN     BIG01R(641)NORTH ERROR PARAMETER FROM DCU ERROR MODEL
BPSA    BIG01R(642)TRANS. FUNCTION -YAW ANTENNA ANGLE
BTHA    BIG01R(643)TRANS. FUNCTION -PITCH ANTENNA ANGLE
CP      BIG01R(644)UPDATED INERTIAL POSITION-N, E, AND D(PZ)
D       BIG01R(647)ERROR IN POSITION-NORTH, EAST, AND DOWN
DTACQ   BIG01R(650)TIME SINCE START OF DCU ACQUISITION ATTEMPT
EPSBGR  BIG01R(651)MISSILE BODY RATE MEASUREMENT BIAS ERROR
EPSCER  BIG01R(654)PU GIMBAL BIASES FOR COSINE RESOLVERS
EPSCFR  BIG01R(657)FRACTIONAL ERROR IN PU COSINE RESOLVER MEASURE
EPSFGR  BIG01R(660)FRACTIONAL ERROR IN MISSILE BODY RATE MEASURE
EPSSER  BIG01R(663)PU GIMBAL ANGLE BIASES FOR SINE RESOLVERS
EPSSFR  BIG01R(666)FRACTIONAL ERROR IN PU SINE RESOLVER MEASURE
ERE     BIG01R(669)DCU ERROR EAST
ERN     BIG01R(670)DCU ERROR NORTH
KPHA    BIG01R(674)TRANSFER FUNCTION CONSTANT-ROLL ANT. ANGLE
KPSA    BIG01R(675)TRANSFER FUNCTION CONSTANT-YAW ANT. ANGLE
KTHA    BIG01R(676)TRANSFER FUNCTION CONSTANT-PITCH ANT. ANGLE
MQ      BIG01R(678)MATCH QUALITY
P1      BIG01R(680)FIRST STAGE ROLL RATE
P2      BIG01R(681)SECOND STAGE ROLL RATE
P3      BIG01R(682)SINGLE STAGE ROLL RATE
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TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

P4	BIG01R(683)RV EP/AB/SE ROLL RATE
P5	BIG01R(684)RV EP ROLL RATE
P6	BIG01R(685)RV AB/SB ROLL RATE
PHC	BIG01R(686)SAU ROLL DRIVE
PSC	BIG01R(687)SAU YAW DRIVE
Q1	BIG01R(688)FIRST STAGE PITCH RATE
Q2	BIG01R(689)SECOND STAGE PITCH RATE
Q3	BIG01R(690)SINGLE STAGE PITCH RATE
Q4	BIG01R(691)RV EP/AB/SE PITCH RATE
Q5	BIG01R(692)RV EP PITCH RATE
Q6	BIG01R(693)RV AB/SB PITCH RATE
R1	BIG01R(694)FIRST STAGE YAW RATE
R2	BIG01R(695)SECOND STAGE YAW RATE
R3	BIG01R(696)SINGLE STAGE YAW RATE
R4	BIG01R(697)RV EP/AB/SE YAW RATE
R5	BIG01R(698)RV EP YAW RATE
R6	BIG01R(699)RV AB/SB YAW RATE
RA	BIG01R(700)EULER ANGLES-ANTENNA TO MISSILE FRAME
RCAPH	BIG01R(703)COSINE ANTENNA PITCH
RCARL	BIG01R(704)COSINE ANTENNA ROLL
RCAYW	BIG01R(705)COSINE ANTENNA YAW
RSAPH	BIG01R(707)SINE ANTENNA PITCH
RSARL	BIG01R(708)SINE ANTENNA ROLL
RSAYW	BIG01R(709)SINE ANTENNA YAW
SIGRGR	BIG01R(710)STANDARD DEVIATION OF MISSILE BODY RATE MEASURE
SIGPRA	BIG01R(711)STANDARD DEVIATION OF RJ RESOLVER MEASUREMENTS
THC	BIG01R(712)SAU PITCH DRIVE
TMXACQ	BIG01R(713)MAXIMUM TIME FOR ACQUISITION BY DCU IN ALTITUDE
TRUON	BIG01R(714)TIME AT WHICH RADAR UNIT TURNED ON
CAPON	BIG01R(715)BASIC AXIAL FORCE COEFFICIENT (POWER ON)
CPM	BIG01R(716)BASIC PITCHING MOMENT COEFFICIENT
CPMQ	BIG01R(717)PITCHING MOMENT Q DAMPING DERIVATIVE
CRM	BIG01R(718)BASIC ROLLING MOMENT COEFFICIENT
CRMP	BIG01R(719)BASIC ROLLING MOMENT DAMPING DERIVATIVE
CYM	BIG01R(720)BASIC YAWING MOMENT COEFFICIENT
CYMR	BIG01R(721)YAWING MOMENT R DAMPING DERIVATIVE
DCAAE	BIG01R(722)INCREMENTAL CA DUE TO AEROELAS. EFFECTS
DCADP	BIG01R(723)INCREMENTAL CA DUE TO DELTAP VANE DEFL.
DCADQ	BIG01R(724)INCREMENTAL CA DUE TO DELTAQ VANE DEFL.
DCADR	BIG01R(725)INCREMENTAL CA DUE TO DELTAR VANE DEFL.
DCAPCF	BIG01R(726)ADDITIVE BASE PRESSURE AXIAL FORCE COEF.
DCASF	BIG01R(727)SKIN FRICTION AXIAL FORCE COEFFICIENT
DCAVM	BIG01R(728)INCREMENTAL CA DUE TO VANE MISALIGNMENT
DCNAE	BIG01R(729)INCREMENTAL CN DUE TO AEROELAS. EFFECTS
DCNDP	BIG01R(730)INCREMENTAL CN DUE TO DELTAP VANE DEFL.
DCNDQ	BIG01R(731)INCREMENTAL CN DUE TO DELTAQ VANE DEFL.
DCNDR	BIG01R(732)INCREMENTAL CN DUE TO DELTAR VANE DEFL.
DCNVM	BIG01R(733)INCREMENTAL CN DUE TO VANE MISALIGNMENT
DCPMAE	BIG01R(734)INCREMENTAL CPM DUE TO AEROELAS. EFFECTS
DCPMDP	BIG01R(735)INCREMENTAL CPM DUE TO DELTAP VANE DEFL.
DCPMDQ	BIG01R(736)INCREMENTAL CPM DUE TO DELTAQ VANE DEFL.

TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

DCPMDR	BIG01R(737)	INCREMENTAL CPM DUE TO DELTAR VANE DEFL.
DCPMVM	BIG01R(738)	INCREMENTAL CPM DUE TO VANE MISALIGNMENT
DCRMAE	BIG01R(739)	INCREMENTAL CRM DUE TO AEROELAS. EFFECTS
DCRMDP	BIG01R(740)	INCREMENTAL CRM DUE TO DELTAP VANE DEFL.
DCRMDQ	BIG01R(741)	INCREMENTAL CRM DUE TO DELTAQ VANE DEFL.
DCRMDR	BIG01R(742)	INCREMENTAL CRM DUE TO DELTAR VANE DEFL.
DCRMVM	BIG01R(743)	INCREMENTAL CRM DUE TO VANE MISALIGNMENT
DCYAE	BIG01R(744)	INCREMENTAL CY DUE TO AEROELAS. EFFECTS
DCYDP	BIG01R(745)	INCREMENTAL CY DUE TO DELTAP VANE DEFL.
DCYDQ	BIG01R(746)	INCREMENTAL CY DUE TO DELTAQ VANE DEFL.
DCYDR	BIG01R(747)	INCREMENTAL CY DUE TO DELTAR VANE DEFL.
DCYMAE	BIG01R(748)	INCREMENTAL CYM DUE TO AEROELAS. EFFECTS
DCYMDP	BIG01R(749)	INCREMENTAL CYM DUE TO DELTAP VANE DEFL.
DCYMDQ	BIG01R(750)	INCREMENTAL CYM DUE TO DELTAQ VANE DEFL.
DCYMDR	BIG01R(751)	INCREMENTAL CYM DUE TO DELTAR VANE DEFL.
DCYMVM	BIG01R(752)	INCREMENTAL CYM DUE TO VANE MISALIGNMENT
GAME	BIG01R(757)	EARTH-RELATIVE FLIGHT ANGLE WRT HORIZONTAL
GAMT	BIG01R(758)	ELEVATION OF LOS TO TARGET WRT LOCAL HOR.
GAMX	BIG01R(759)	ELEVATION MISSILE X-AXIS WRT LOCAL HOR.
GRL	BIG01R(760)	GROUND RANGE LAUNCH POINT TO CURRENT POS.
GRT	BIG01R(761)	GROUND RANGE FROM CURRENT POSITION TO TARGET
NEDDOT	BIG01R(763)	EARTH-RELATIVE VELOCITY IN LOCAL NED FRAME
SIGE	BIG01R(771)	EARTH-RELATIVE FLIGHT PATH ANGLE WRT NORTH
SIGT	BIG01R(772)	AZIMUTH OF LINE-OF-SIGHT TO TARGET
SIGX	BIG01R(773)	AZIMUTH OF MISSILE X AXIS
SRT	BIG01R(774)	SLANT RANGE TO TARGET
URELM	BIG01R(775)	
VRELM	BIG01R(780)	VEL. OF MISSILE RELATIVE TO AIR IN MISSILE
XD	BIG01R(784)	MISSILE POS. VECTOR IN DESIRED (TARGET) FRAME
XEDOT	BIG01R(787)	MISSILE EARTH-RELATIVE VELOCITY IN EARTH FRAME
XTME	BIG01R(790)	POSITION OF MISSILE RELATIVE TO TARGET IN E
DC	BIG01R(818)	FIXED DRIFTS-X ,Y, AND Z
DLV	BIG01R(821)	DELTA MAGNITUDE OF INERTIAL VELOCITY
DSMAT	BIG01R(824)	G-SENSITIVE DRIFT PARTIAL DERIVATIVE MATRIX
FDIMS	BIG01R(833)	IMS FIXED DRIFT RATE
NSF	BIG01R(836)	IMS NEGATIVE SCALE FACTORS-X, Y, AND Z
PSF	BIG01R(839)	IMS POSITIVE SCALE FACTORS-X, Y, AND Z
RSDV	BIG01R(842)	IMS RESIDUAL DELTA V MEASUREMENT
RYX	BIG01R(845)	IMS AXIS NONORTHOGONALITY-Y TO X
RZX	BIG01R(846)	IMS AXIS NONORTHOGONALITY-Z TO X
RZY	BIG01R(847)	IMS AXIS NONORTHOGONALITY-Z TO Y
TQMAG	BIG01R(848)	MAGNITUDE OF IMS TORQUER RATE
IXX	BIG01R(849)	
IYY	BIG01R(850)	
IZZ	BIG01R(851)	
MOLD	BIG01R(852)	
NOZDLX	BIG01R(853)	NOZZLE MOMENT ARM COMPONENT ALONG X AXIS
TRCSON	BIG01R(855)	TIME AT WHICH RCS JET WAS TURNED ON
EAD	BIG01R(863)	EULER ANGLE RATES-ANTENNA TO MISSILE FRAME
EADD	BIG01R(866)	EULER ANGLE ACCS.-ANTENNA TO MISSILE FRAME
TACQ	BIG01R(870)	TIME REQUIRED FOR DCU ACQUISITION

TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

```
TOCPI  BIG01R(371)TIME AT WHICH CPI COMMAND WAS RECEIVED
FSIFLG BIG01I( 58)FIRST STAGE IGNITION FLAG
FSSFLG BIG01I( 59)FIRST STAGE SEPARATION FLAG
ID1    BIG01I( 67) 0   INPUT DISCRETE WORD 1
ID2    BIG01I( 68) 0   INPUT DISCRETE WORD 2
ID3    BIG01I( 69) 0   INPUT DISCRETE WORD 3
IDER   BIG01I( 70)DERIVATIVE COMPUTATION COUNTER
INITFL BIG01I( 71)INITIALIZATION FLAG
ISEQ   BIG01I( 72)FLIGHT SEQUENCE INDICATOR
ISTAGE BIG01I( 73)STAGE FLAG;=1 LAUNCH AND STAGE;=0 OTHERWISE
OD2    BIG01I( 98) 0   OUTPUT DISCRETE WORD 2
RVSFLG BIG01I(110)RV SEPARATION FLAG
SSIFLG BIG01I(113)SECOND STAGE IGNITION FLAG
SSSFLG BIG01I(114)SECOND STAGE SEPARATION FLAG
THROPT BIG01I(113)THRUST OPTION (1-USE TSLV;  2-USE TVAC)
JDFLG  BIG01I(213)JET FLAG (0=NO JD;  1-JD)
MSTOPT BIG01I(227)MASS TABLE OPTION FLAG
RVJ    BIG01I(412)REACTION JET-ON FLAGS (3 JETS)
ARMCOD BIG01I(435)CHEFU ARM CODE
BACODE BIG01I(436)BATTERY ACTIVATE CODE
BAFLG  BIG01I(437)BATTERY ACTIVATED FLAG
COWRD  BIG01I(438)SAF REQUIRED CUTOFF WORD
EPWRD  BIG01I(439)SAF REQUIRED EP WORD
FIRCOD BIG01I(440)CHEFU FIRE CODE
FSAAFL BIG01I(441)FIRST STAGE IGNITION S AND A ARM FLAG
FSACOD BIG01I(442)FIRST STAGE IGNITION S AND A CODE
IBCFL  BIG01I(443)BOOST CONFIG. (1-SINGLE,  2-TWO STAGE)
IFAWRD BIG01I(444)SAF REQUIRED INFLIGHT ARMING WORD
LOWRD  BIG01I(445)SAF REQUIRED LIFT-OFF WORD
OD1    BIG01I(446) 0   OUTPUT DISCRETE WORD 1
PACNF  BIG01I(447)PAC NO FAULT FLAG
RVSWRD BIG01I(448)SAF REQUIRED RV SEPARATION WORD
TSIWRD BIG01I(449)SAF REQUIRED TERMINAL SEQUENCE INITIATE WORD
WHAFLG BIG01I(450)WARHEAD ARMED FLAG
WHBFLG BIG01I(451)WARHEAD BURST FLAG
WHT    BIG01I(452)WARHEAD OPTION
CLRDV  BIG01I(463)CLEAR IMS DELTA-V COUNTERS FLAG
DAYSC  BIG01I(464)NUMBER OF DAYS SINCE LAST IMS CALIBRATION
ID4    BIG01I(515) 0   INPUT DISCRETE WORD 4
NDLV   BIG01I(559)NEGATIVE PULSE COUNT-X, Y, AND Z
OD4    BIG01I(592) 0   OUTPUT DISCRETE WORD 4
PDLV   BIG01I(593)POSITIVE PULSE COUNT-X, Y, AND Z
TRQTST BIG01I(625)IMS TORQUER BITE TEST FLAG
ID5    BIG01I(671) 0   INPUT DISCRETE WORD 5
ID6    BIG01I(672) 0   INPUT DISCRETE WORD 6
INITDE BIG01I(673)DCU ERROR MODEL INITIALIZATION FLAG
MNFCNT BIG01I(677)MAJOR FRAME COUNTER
OD5    BIG01I(679) 0   OUTPUT DISCRETE WORD 5
RFDAFL BIG01I(705)FIX DATA READ BY PACP-FLAG
ANTON  BIG01I(754)ANTENNA ON FLAG-STABILIZE ANTENNA RECEIVED
DCLON  BIG01I(755)DCL ON FLAG
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TABLE I-1. TRW PROGRAM OUTPUT VARIABLES (Continued)

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FHYDON BIG01I(756)FIRST STAGE HYDRAULIC SYSTEM ON FLAG
NAVFLG BIG01I(762)BEGIN NAVIGATION FLAG
OD3    BIG01I(766) 0   OUTPUT DISCRETE WORD 3
PWRFLG BIG01I(767)MISSILE POWER-ON FLAG
RCSON  BIG01I(768)REACTION CONTROL SYSTEM ON FLAG
RUON   BIG01I(769)RADAR UNIT ON FLAG
SHYDON BIG01I(770)SECOND STAGE HYDRAULIC SYSTEM ON FLAG
VCSGG1 BIG01I(778)VCS GAS GENERATOR NUMBER 1 INITIATE FLAG
VCSGG2 BIG01I(779)VCS GAS GENERATOR NUMBER 2 INITIATE FLAG
WNDOPT BIG01I(783)WIND OPTION; 0-NO WINDS, 1-NEV, 2-HAV
BAWRD  BIG01I(793)BATTERY ACTIVATE WORD FROM PACP
COFLG  BIG01I(794)SAF CUTOFF FLAG
EPFLG  BIG01I(795)SAF EARTH PENETRATOR FLAG
FSAWRD BIG01I(796)FIRST STAGE IGNITION S AND A WORD FROM PACP
FSIAFL BIG01I(797)FIRST STAGE IGNITION ARMED FLAG
FSIWRD BIG01I(798)FIRST STAGE IGNITION WORD FROM PACP
FSSAFL BIG01I(799)FIRST STAGE SEPARATION ARMED FLAG
FSSWRD BIG01I(800)FIRST STAGE SEPARATION WORD FROM PACP
IFAFLG BIG01I(801)SAF INFLIGHT ARMING FLAG
LOFLG  BIG01I(802)SAF LIFT-OFF FLAG
NB     BIG01I(803)NUMBER OF BITS EXPECTED IN CURRENT SAF WORD
NBAB   BIG01I(804)NUMBER OF BATTERY ACTIVATE BITS ACCUMULATED
NFSAB  BIG01I(805)NUMBER OF FSI S AND A BITS ACCUMULATED
NFSIB  BIG01I(806)NUMBER OF FIRST STAGE IGNITION BITS ACCUMULATED
NFSSB  BIG01I(807)NUMBER OF FIRST STAGE SEP. BITS ACCUMULATED
NSAFB  BIG01I(808)NUMBER OF SAF BITS ACCUMULATED
NSSIB  BIG01I(809)NUMBER OF BITS ACCUMULATED IN SSI WORD
NSSSB  BIG01I(810)NUMBER OF SECOND STAGE SEP. BITS ACCUMULATED
SAFRVS BIG01I(811)SAF RV SEP. FLAG
SAFWRD BIG01I(812)CURRENT SAF WORD FROM PACP
SSIAFL BIG01I(813)SECOND STAGE IGNITION ARMED FLAG
SSIWRD BIG01I(814)SECOND STAGE IGNITION WORD RECEIVED FROM PACP
SSSAFL BIG01I(815)SECOND STAGE SEPARATION ARMED FLAG
SSSWRD BIG01I(816)SECOND STAGE SEPARATION WORD FROM PACP
TSIFLG BIG01I(817)SAF TERMINAL SEQUENCE INITIATE FLAG
OD2P   BIG01I(854) 0   PREVIOUS VALUE OF OUTPUT DISCRETE WORD 2
ACQFLG BIG01I(851)DCU ACQUISITION FLAG
OD5P   BIG01I(862) 0   PREVIOUS VALUE OF OUTPUT DISCRETE WORD 5
SCNFLG BIG01I(859)SCAN MODE FLAG
R      BIG01D( 54)DISTANCE FROM EARTH'S CENTER
SIMTIM BIG01D( 56)SIMULATION TIME FROM START OF NAVIGATION
TDER   BIG01D( 58)TIME AT WHICH DERIVATIVES ARE COMPUTED
X      BIG01D(129)STATE VECTOR-XE, PSI, U, AND P VECTORS
XP     BIG01D(143)PSEUDO (INTERMEDIATE)STATE VECTOR
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